

Security of the Fischlin Transform in Quantum Random Oracle Model

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Abstract. The Fischlin transform yields non-interactive zero-knowledge proofs with straight-line extractability in the classical random oracle model. This is done by forcing a prover to generate multiple accepting transcripts through a proof-of-work mechanism. Whether the Fischlin transform is straight-line extractable against quantum adversaries has remained open due to the difficulty of reasoning about the *likelihood* of query transcripts in the quantum-accessible random oracle model (QROM), even when using the compressed oracle methodology. In this work, we prove that the Fischlin transform remains straight-line extractable in the QROM, via an extractor based on the compressed oracle. This establishes the post-quantum security of the Fischlin transform, providing a post-quantum straight-line extractable NIZK alternative to Pass’ transform with smaller proof size. Our techniques include tail bounds for sums of independent random variables and for martingales as well as symmetrization, query amplitude and quantum union bound arguments.

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1 Introduction

Non-interactive zero-knowledge proofs (NIZKs) obtained from interactive protocols via the Fiat–Shamir paradigm form a central tool in modern cryptography. Beyond zero-knowledge, many applications require *straight-line extractability*, namely the ability to efficiently extract a witness from any prover that convinces the verifier, without rewinding. The Fischlin transform [Fis05] provides a prominent approach for achieving straight-line extractable NIZKs from Σ -protocols in the random oracle model. It constitutes an alternative to Pass’ transform [Pas03], replacing the use of commitments by a rejection-sampling mechanism based on oracle queries. As a result, the Fischlin transform yields small non-interactive proofs with strong extraction guarantees. The key constructive element in the Fischlin transform is what would now be called a proof-of-work-like mechanism. The challenges that are chosen uniformly at random by the verifier in the interactive setting can now be chosen freely by the prover, but need to fulfil a hash condition, e.g., the hash of the transcript needs to start with l zeroes. Modeling the hash function as a random oracle now implicitly randomizes the challenge as the pre-image of the hash predicate is random.

With the advent of quantum computing, it has become necessary to analyze the security of random-oracle-based constructions in the presence of quantum adversaries. This has led to the development of the quantum random oracle

model (QROM) [BDF⁺11], in which adversaries may query the random oracle in superposition. Over the past decade, a substantial body of work has established a versatile technical toolbox for proving the post-quantum security of classical constructions in the QROM. These techniques have been successfully applied to a wide range of Fiat–Shamir-type transforms, including Pass’ transform [CMS19,DFMS22,RT25] and related primitives. However, despite its importance, the Fischlin transform has so far resisted security analysis in the QROM. Existing proof techniques do not readily extend to this setting, and no QROM extractability proof for the Fischlin transform was previously known.

The main difficulty in analyzing the Fischlin transform in the QROM stems from the structure of its extraction mechanism. Informally, a prover succeeds only if it finds an oracle query whose output satisfies a stringent acceptance condition, which in turn requires many attempts, except with tiny probability. A classical extractor can thus read off two accepting transcripts from the adversarial prover’s oracle query inputs, allowing extraction of a witness via special soundness. In the quantum setting, however, this reasoning becomes problematic. The adversary’s output depends only on the single accepting query, the unsuccessful ones are in some sense *counterfactual*. This stands in sharp contrast to Pass’ transform, where accepting transcripts contain explicit commitments to responses for all challenges that can be leveraged for extraction even in the quantum setting. As a result, proving that a quantum adversary must invest substantial query complexity in order to succeed requires new techniques.

Our results. In this work, we prove the straight-line extractability of the Fischlin transform in the QROM. Our result provides concrete bounds for a wide range of parameter choices that include an infinite family of parameter sets for asymptotic security. In more detail, we have the following

Theorem 1 (Informal version of Corollary 1). *Let Σ be a Σ -protocol for a witness relation R with special soundness and unique responses. Under suitable conditions on the parameters of the Fischlin transform that allow unbounded values for the parallel repetition parameter of the Fischlin transform, k , the Fischlin transform $\text{Fis}[\Sigma]$ is a proof of knowledge with straight-line extractability. The simulation can be made perfect and the extractor succeeds whenever a q -query malicious prover does, except with probability no more than*

$$\varepsilon_{\text{ex}} \leq q^2 \cdot \text{negl}(k).$$

Technical Overview. Defining the straight-line extractor for the Fischlin transform is straightforward. We simulate the random oracle towards the prover using a compressed oracle [Zha19]. Once the prover has finished and its proof is verified, the extractor measures the compressed oracle database register. If the measurement result contains two accepting transcripts for the same repetition with different challenges, the special soundness extractor of the Σ -protocol is used to produce a witness.

The main work goes into proving that no prover can succeed with significant probability at outputting a proof that involves a fixed commitment vector chosen ahead of time (“deterministic prover”) so that the extractor still fails. This result is then lifted to a general prover that chooses the commitment vector in a randomized manner using an intricate but compact query amplitude argument using a quantum union bound.

To show the result for deterministic provers, our strategy is reminiscent of the Monty Hall paradox. Here, a game master presents three doors. Behind a random one is a car, while there are goats behind the other two. The participant can now choose a door, which is, however, not opened. The game master now opens a *different* door which reveals a goat. The participant can now decide: stick to their choice of door, or choose the only remaining door instead. The optimal strategy is to switch, which reveals the car with probability $2/3$. What makes this situation appear paradoxical is the fallacy of regarding the remaining door as a fixed object instead of a random variable. The game master chooses the door they open depending on the randomness of the placement of the car, which correlates the car’s placement with the remaining doors.

When a prover outputs a valid Fischlin proof, a number of hash values are now known (to evaluate to 0). This can only be achieved by correlating the choice of challenges with the random oracle. Clearly this can only happen if the probability that some hash outputs not involved in proof verification are 0 is now *less* than suggested by the uniform distribution, otherwise the oracle output distribution would be biased. In the compressed oracle, this means that the oracle registers corresponding to valid transcripts not involved in the verification of the proof output by the adversarial prover cannot contain the uniform superposition of outputs anymore. This deviation is then exploited by the extractor.

From a technical perspective, the task of making this argument rigorous is formidable. The output length of the random oracle is logarithmic, preventing the usual tight relationship between oracle register content and adversarial view up to negligible error. We therefore need to explicitly analyze every switch between the “compressed” and “uncompressed” bases of the superposition oracle using Chernoff bounds. The high-level strategy is now as follows. Pick a number η such that (i) it is unlikely that there are η oracle inputs in a certain sub-domain that map to zero (by an additional standard Chernoff bound), but (ii) it is likely to find roughly $\eta - k$ zeroes, where k is the number of parallel repetitions in the Fischlin transform and thus the number of hash values that need to be zero for a Fischlin proof to be accepted. We then use the fact that it is unlikely to find η zeroes, but likely to find η zeroes given that the prover has succeeded and the extractor has failed, to conclude that the probability that the prover succeeds and the extractor fails is small.

Proving that it is likely to find η zeroes given that the prover has succeeded and the extractor has failed is non-trivial. Conditioning (or rather projecting) on these events renders the oracle register non-uniform and concentration inequalities for independent Bernoulli random variables no longer apply. Fortunately, symmetrizing the state and using properties of the “extractor fails” projector, we

can define a suitable *martingale* based on the outcomes of measuring the different database registers and apply the Azuma-Hoeffding concentration inequality to get the desired bound.

Overall, our analysis establishes that, even in the presence of quantum superposition queries, success in the Fischlin transform implicitly certifies the existence of many “hidden” accepting transcripts. While these transcripts may not be classically observable, they can be detected through concentration phenomena in the prover’s measurement process, enabling straight-line extraction in the QROM.

1.1 Organization

Section 2 reviews the QROM and the compressed oracle framework. Section 3 proves straight-line extractability of the Fischlin transform in the QROM. Section 4 establishes zero-knowledge.

2 Preliminaries

For asymptotic security analysis, protocol parameters are expressed as a function of the security parameter $\lambda \in \mathbb{N}$. A (possibly negative) function $f(\lambda)$ is called negligible if $|f(\lambda)| \leq 1/p(\lambda)$ for any polynomial $p(\lambda)$ and all sufficiently large λ . We use negl to denote a negligible function. A function which is not negligible is called noticeable. For any witness relation $R \subset \mathcal{X} \times \mathcal{W}$ we call the elements of $\mathcal{X}$ instances, the elements of $\mathcal{W}$ witnesses, and define the associated language $\mathcal{L}_R = \{x \in \mathcal{X} \mid \exists w \in \mathcal{W} : (x, w) \in R\}$. We denote by $\delta(\cdot, \cdot)$ the maximal 1-sample computational distinguishing advantage between two probability distributions. We allow algorithms as arguments here, which indicate computational distinguishability of their outputs.

A common idealization is the random oracle model where parties have access to a random function H with some domain and range depending on λ . Our random oracles have output length at most polynomial in the security parameter.

Definition 1 (Σ -Protocol). A Σ -protocol $\Sigma = (\mathcal{P}, \mathcal{V})$ for a relation $R \subseteq \mathcal{X} \times \mathcal{W}$ is a three-round two-party interactive protocol between a Prover $\mathcal{P} = (\mathcal{P}_1, \mathcal{P}_2)$ and a verifier $\mathcal{V}$ where the prover produces $(a, \text{st}) \leftarrow \mathcal{P}_1(x, w)$ and sends the commitment $a \in \mathcal{M}$ to the verifier, the verifier sends a random challenge $c \leftarrow \mathcal{C}$ (public-coin) and the prover sends a response $z \leftarrow \mathcal{P}_2(\text{st}, c)$. Finally, the verifier outputs the verdict $b \leftarrow \mathcal{V}(x, a, c, z)$, with $b = 1$ meaning “valid”. We denote the part of the interaction that produces the communication transcript as $(a, c, z) \leftarrow \langle \mathcal{P}(x, w), \mathcal{V} \rangle$. It has to satisfy the following properties:

1. *Completeness.* For any $(x, w) \in R$, any $(a, c, z) \leftarrow \langle \mathcal{P}(x, w), \mathcal{V} \rangle$ it holds $\mathcal{V}(x, a, c, z) = 1$.
2. *Special Soundness.* There exists a probabilistic polynomial-time (PPT) extractor algorithm Ext_{ss} such that for all $x \in \mathcal{X}$ and for all pairs of transcripts (a, c, z) and (a, c', z') with $c \neq c'$ such that $\mathcal{V}(x, a, c, z) = \mathcal{V}(x, a, c', z') = 1$,

the extractor produces a witness $w \leftarrow \text{Ext}_{\text{ss}}(x, a, c, z, c', z')$ satisfying $(x, w) \in R$.

Definition 2 (Min-Entropy). The *min-entropy* of a discrete random variable X over a finite set $\mathcal{X}$ is defined as:

$$H_{\infty}(X) = -\log_2 \left(\max_{x \in \mathcal{X}} \Pr[X = x] \right)$$

where $\Pr[X = x]$ denotes the probability that X takes the value x .

We now list some additional properties required by our Σ -protocol the make the transformation from interactive to non-interactive, work:

Definition 3 (Commitment Entropy). Let Σ be a Σ -protocol. We say Σ has commitment entropy $\alpha \in \mathbb{R}_{\geq 0}$ if, for any $(x, w) \in R$ and $a \leftarrow \mathcal{P}_1(x, w)$ it holds that $H_{\infty}(a) \geq \alpha$.

Definition 4 (Unique Responses). Let Σ be a Σ -protocol. We say Σ has unique responses, if for all tuples $(x, a, c, z, z') \leftarrow A(\lambda)$ with $z \neq z'$, either $\mathcal{V}_1(x, a, c, z) = 0$ or $\mathcal{V}_1(x, a, c, z') = 0$.

Definition 5 (Honest-Verifier Zero-Knowledge (HVZK)). Let Σ be a Σ -protocol for relation R . We say Σ has honest-verifier zero-knowledge (HVZK) if there exists a PPT algorithm Sim_{HVZK} , the zero-knowledge simulator, such that for any quantum polynomial-time (QPT) algorithm $\mathcal{D} = (\mathcal{D}_0, \mathcal{D}_1)$, it holds that

$$\left| \Pr_{\substack{(x, w, \text{st}) \leftarrow \mathcal{D}_0(1^\lambda) \\ (a, c, z) \leftarrow \langle \mathcal{P}(x, w), \mathcal{V} \rangle}} [1 \leftarrow \mathcal{D}_1(a, c, z, \text{st}) \wedge (x, w) \in R] \right. \\ \left. - \Pr_{\substack{(x, w, \text{st}) \leftarrow \mathcal{D}_0(1^\lambda) \\ (a, c, z) \leftarrow \text{Sim}_{\text{HVZK}}(x)}} [1 \leftarrow \mathcal{D}_1(a, c, z, \text{st}) \wedge (x, w) \in R] \right| \leq \text{negl}(\lambda).$$

Below we sometimes use a stronger zero-knowledge property where the simulator is able to generate proofs for a specific challenge.

Definition 6 (Special Honest Verifier Zero-Knowledge). A Σ -protocol Σ is special honest-verifier zero-knowledge (SHVZK) if it is HVZK and there exists a simulator $\text{Sim}_{\text{SHVZK}}$ such that $\text{Sim}_{\text{SHVZK}}(x)$ simulates transcripts (a, c, z) by sampling $c \leftarrow \mathcal{C}$ and running $(a, z) \leftarrow \text{Sim}_{\text{SHVZK}}(x, c)$.

We now formally describe a Non-Interactive Zero-Knowledge(NIZK) proof with online extractors in the Random Oracle Model.

Definition 7 (Non-Interactive Zero-Knowledge proof (NIZK)). A pair $(\mathcal{P}, \mathcal{V})$ of probabilistic polynomial-time algorithms is called a non-interactive zero-knowledge proof (NIZK) for relation R if the following holds:

Completeness. For any $(x, w) \in R$ and any $\pi \leftarrow \mathcal{P}^H(x, w)$ we have $\Pr[\mathcal{V}^H(x, \pi) = 1] = 1 - \text{negl}(\lambda)$.

Zero-Knowledge. *There exists a probabilistic polynomial-time simulator Sim such that for any quantum polynomial-time distinguisher $\mathcal{D} = (\mathcal{D}_0, \mathcal{D}_1)$, we have*

$$\left| \Pr \left[\mathcal{D}_1^H(\pi, \text{st}) = 1 : (x, w, \text{st}) \leftarrow \mathcal{D}_0^H(1^\lambda), (x, w) \in R, \pi \leftarrow \mathcal{P}^H(x, w) \right] - \Pr \left[\mathcal{D}_1^{H'}(\pi, \text{st}) = 1 : (x, w, \text{st}) \leftarrow \mathcal{D}_0^H(1^\lambda), (x, w) \in R, (H', \pi) \leftarrow \text{Sim}(x) \right] \right| \leq \text{negl}(\lambda).$$

Here, the simulator outputting H' is to be understood as outputting (reprogramming) instructions how to instantiate H' relative to H .

In the QROM, straight-line extractability can be defined as follows, following [DFMS22]. Let $\mathcal{P}^*$ be a dishonest prover that potentially outputs some additional auxiliary (possibly quantum) output Z next to π . We then consider an interactive algorithm Ext , called *online extractor*, which takes the security parameter λ as input and simulates the answers to the oracle queries in the execution of $\mathcal{V}^H \circ \mathcal{P}^{*H}(x)$, which we define to run $(\pi, Z) \leftarrow \mathcal{P}^{*H}(x)$ followed by $v \leftarrow \mathcal{V}^H(x, \pi)$; furthermore, at the end, Ext outputs $w \in \mathcal{W}$. We denote the execution of $\mathcal{V}^H \circ \mathcal{P}^{*H}(x)$ with the calls to H simulated by Ext , and considering Ext 's final output w as well, as $(\pi, Z; v; w) \leftarrow \mathcal{V}^{\text{Ext}} \circ \mathcal{P}^{*\text{Ext}}(x)$.

Definition 8 (Non-adaptive variant of Definition 3.1 in [DFMS22]). *A NIZK in the (QROM) for relation $R \subset \mathcal{X} \times \mathcal{W}$ is a proof of knowledge with straight-line extractability against non-adaptive adversaries if there exists an online extractor Ext , and functions ε_{sim} (the simulation error) and ε_{ex} (the extraction error), with the following properties. For any $\lambda \in \mathbb{N}$, for any $x \in \mathcal{X}$ and for any dishonest prover $\mathcal{P}^*$ making no more than q queries,*

$$\delta([(\pi, Z, v)]_{\mathcal{V}^H \circ \mathcal{P}^{*H}(x)}, [(\pi, Z, v)]_{\mathcal{V}^{\text{Ext}} \circ \mathcal{P}^{*\text{Ext}}(x)}) \leq \varepsilon_{\text{sim}}(\lambda, q)$$

and

$$\Pr[v = \text{accept} \wedge (x, w) \notin R : (\pi, Z; v; w) \leftarrow \mathcal{V}^{\text{Ext}} \circ \mathcal{P}^{*\text{Ext}}(x)] \leq \varepsilon_{\text{ex}}(\lambda, q).$$

Furthermore, the runtime of Ext is polynomial in $\lambda + q$, and $\varepsilon_{\text{sim}}(\lambda, q)$ and $\varepsilon_{\text{ex}}(\lambda, q)$ are negligible in λ whenever q is polynomial in λ .

The Fischlin transform.

We use the version of the Fischlin transform described in [Ks22].

Definition 9 (The Fischlin transform). *The security parameter λ defines the integers k, ℓ, t , related as $k \cdot \ell = \lambda$, and $t = \lceil \log \lambda \rceil \cdot \ell$. The protocol uses a hash function $H : \mathbf{X} = \{(\mathbf{a}, i, c_i, z_i)\} = X_{\mathbf{a}} \times X_i \times X_{c_i} \times X_{z_i} \mapsto \mathcal{Y} = \{0, 1\}^\ell$, modeled as a random oracle, and a Σ -protocol $\Sigma = ((\mathcal{P}_{\Sigma,1}, \mathcal{P}_{\Sigma,2}), \mathcal{V}_\Sigma)$. The prover and verifier of the Fischlin transform $\text{Fis}[\Sigma]$ are defined as follows.*

$\mathcal{P}^{\text{Fis}[\Sigma]}(x, w) :$

1. For each $i \in [k]$, compute $(a_i, \text{state}_i) \leftarrow \mathcal{P}_{\Sigma,1}(x, w)$
2. Set $\mathbf{a} = (a_i)_{i \in [k]}$, and initialize $c_i = -1$ for each $i \in [k]$
3. For each $i \in [k]$, do the following:

- (a) If $c_i > t$, abort. Otherwise increment c_i and compute $z_i = \mathcal{P}_{\Sigma,2}(\text{state}_i, c_i)$
(b) If $H(\mathbf{a}, i, c_i, z_i) \neq 0^\ell$, repeat Step 3a
4. Output $\pi = (\mathbf{a}, \mathbf{c}, \mathbf{z})$, where $\mathbf{c} = (c_i)_{i \in [k]}$ and $\mathbf{z}$ are defined analogously.
 $\mathcal{V}^{\text{Fis}[\Sigma]}(x, \pi)$:
1. Parse $(\mathbf{a}, \mathbf{c}, \mathbf{z}) = \pi$
2. For each $i \in [k]$, verify that $H(\mathbf{a}, i, c_i, z_i) = 0^\ell$ and $\mathcal{V}_\Sigma(x, (a_i, c_i, z_i)) = 1$, aborting with output 0 if not
3. Accept by outputting 1

2.1 Quantum Random Oracle (QROM) and Compressed Oracle Technique

In this section we summarize the mathematical tools and notational conventions used throughout the paper. We review basic facts about Hilbert spaces and operators, introduce the computational and Fourier bases for finite Abelian groups, and recall the standard encoding of functions into quantum oracles. These foundations are needed to formalize the compressed oracle technique and to analyze the behavior of quantum adversaries making oracle queries.

Hilbert Spaces and Operators. We work over finite-dimensional complex Hilbert spaces. Unless stated otherwise, $\mathcal{H} = \mathbb{C}^d$ for some d , with the usual bracket notation. Linear operators between spaces $\mathcal{H}$ and $\mathcal{H}'$ are denoted $\mathcal{L}(\mathcal{H}, \mathcal{H}')$, with $\mathcal{L}(\mathcal{H})$ for endomorphisms. A (pure) quantum state is a unit vector $|\psi\rangle \in \mathcal{H}$.

Norms. For $A \in \mathcal{L}(\mathcal{H}, \mathcal{H}')$, $\|A\|$ denotes the operator norm. If $\mathcal{H} = \bigoplus_{i=1}^m \mathcal{H}_i$ and A acts as B_i on each $\mathcal{H}_i$, then $\|A\| = \max_i \|B_i\|$. When A is defined only on a subspace, it can be extended by zero outside without ambiguity in its norm.

Any finite set $\mathcal{Y}$ defines an associated quantum register with state space $\mathbb{C}\mathcal{Y}$ computational basis $\{|y\rangle\}_{y \in \mathcal{Y}}$. We will consider the extension $\overline{\mathcal{Y}} = \mathcal{Y} \cup \{\perp\}$ and the superspace $\mathbb{C}\overline{\mathcal{Y}} \supset \mathbb{C}\mathcal{Y}$.

We will denote the function table stored in a quantum register by $|H\rangle = \bigotimes_x |H(x)\rangle \in (\mathbb{C}\mathcal{Y})^{\otimes \mathcal{X}}$. We implicitly consider the different registers to be labeled by $x \in \mathcal{X}$ in the obvious way.

Compressed Oracle Technique. The compressed oracle formalism introduced by Zhandry [Zha19] is a way to simulate a quantum-accessible random oracle in a way that provides additional features for reductions. Instead of storing the full random function, we maintain a *compressed database* D that records only the points queried so far and their outputs. This suffices for simulating any adversary's interaction while simplifying the analysis.

Instead of a uniformly random choice of function, consider a superposition $\frac{1}{\sqrt{|\mathfrak{H}|}} \sum_H |H\rangle$, where $\mathfrak{H}$ is the set of functions H is sampled uniformly from. This is the purified oracle which is indistinguishable from the original random oracle for any (quantum) query algorithm since the queries commute with measuring the superposition. The initial state of the oracle registers is given by

$$|I_{00}\rangle = \frac{1}{\sqrt{|\mathfrak{H}|}} \sum_H |H\rangle = \frac{1}{\sqrt{|\mathfrak{H}|}} \bigotimes_x \left(\sum_y |y\rangle_{D_x} \right) =: \bigotimes_x |+\mathcal{Y}\rangle_{D_x}. \quad (1)$$

Compressed Oracle. The Compressed Oracle¹ is now obtained by re-defining each local register D_x to have state space $\mathbb{C}\overline{\mathcal{Y}}$, embedding the superposition oracle into it, and applying the basis change unitary

$$\text{Comp} = |\perp\rangle\langle +^l| + |+^l\rangle\langle \perp| + (\mathbb{1} - |+^l\rangle\langle +^l| - |\perp\rangle\langle \perp|). \quad (2)$$

This compression operator acts on a single subregister of the compressed oracle database register. As for any operator, we denote by Comp_{D_x} the compression operator applied to a particular register D_x . For the compression operator, we additionally abuse notation slightly by writing

$$\text{Comp}_{D_S} = \bigotimes_{x \in S} \text{Comp}_{D_x}$$

and thus also $\text{Comp}_D = \left(\text{Comp}^{\otimes |\mathcal{X}|} \right)_D$.

The compression operator $\text{Comp}_D := \bigotimes_x \text{Comp}_{D_x}$ maps $|II_0\rangle$ to

$$\begin{aligned} |\Delta_0\rangle &:= \text{Comp} |II_0\rangle = \left(\bigotimes_x \text{Comp}_{D_x} \right) |+_{\mathcal{Y}}^{\mathcal{X}}\rangle \\ &= \bigotimes_x \text{Comp}_{D_x} |+_{\mathcal{Y}}\rangle = \bigotimes_x |\perp\rangle = |\perp^{\mathcal{X}}\rangle, \end{aligned}$$

which is the quantum representation of the trivial database that maps any $x \in \mathcal{X}$ to $\perp$.

As a consequence, the internal state of the compressed oracle after q queries is supported by computational basis states $|D\rangle$ for which $D(x) = \perp$ (respectively $\hat{D}(x) = \perp$) for all but (at most) q choices of x . We call the number of registers that are in a state other than $|\perp\rangle$ the *size* of the database.

An oracle query to the superposition oracle is implemented using the unitary operator O given by

$$O : |\mathbf{x}\rangle_X |y\rangle_Y \otimes |H\rangle_D \mapsto |\mathbf{x}\rangle_X |y \oplus H(x)\rangle_Y \otimes |H\rangle_D. \quad (3)$$

When the compressed oracle is queried, a unitary O_{XYD} , acting on the query registers X and Y and the oracle register D , is applied, given by

$$O_{XYD} = \sum_x |x\rangle\langle x|_X \otimes O_{YD_x}^x,$$

with

$$O_{YD_x}^x = \text{Comp}_{D_x} \text{CNOT}_{YD_x} \text{Comp}_{D_x}$$

where $\text{CNOT}_{YD_x} |y\rangle |y_x\rangle = |y \oplus y_x\rangle |y_x\rangle$ for $y, y_x \in \{0, 1\}^l$ and acts as identity on $|y\rangle |\perp\rangle$

¹ More precisely we are using the *pre-compressed* oracle, the compressed oracle is obtained from the pre-compressed oracle via the standard sparse representation

Query algorithms. We now outline the structure of a quantum algorithm interacting with a compressed oracle. The adversary has registers $XYZO$, where X is the query input, Y the query output, Z is the local workspace of the algorithm and O is the register that is measured to produce the final output. D is the compressed oracle database register that is inaccessible to the algorithm except via oracle queries.

Let $|\phi_i\rangle$ denote the adversary's state before the $(i+1)$ -th query, and $|\phi'_i\rangle$ the state immediately after that query,

$$|\phi'_i\rangle = O_{XYD}|\phi_i\rangle.$$

The adversary then applies a unitary U^2 on (x, u, z) , leaving D unchanged, to obtain

$$|\phi_{i+1}\rangle = U_{XYZO}|\phi'_i\rangle.$$

Thus, after q queries, the overall computation evolves as:

1. Start in some initial state $|\phi_0\rangle$.
2. For each $1 \leq i \leq q$:
 - (a) Apply the oracle O_{XYD} to obtain $|\phi'_i\rangle$.
 - (b) Apply U_{XYZO} to obtain $|\phi_i\rangle$.
3. At the end, perform a computational basis measurement of register O .

2.2 Reprogramming Quantum Random Oracle

We need an adaptive reprogramming lemma from [GHHM21].

Let $\mathcal{X}, \mathcal{X}', \mathcal{Y}$ be finite sets. Consider the following game $\mathbf{Repro}_b$, where $b \in \{0, 1\}$:

1. Sample an initial random oracle $O_0 : \mathcal{X} \rightarrow \mathcal{Y}$.
2. Define O_1 to be O_0 , but with access to a **Reprogram** oracle: on input a probability distribution p on $\mathcal{X} \times \mathcal{X}'$, it samples $(x, x') \leftarrow p$ and $y \leftarrow \mathcal{Y}$, sets $O_1(x) := y$, and returns (x, x') .
3. An adversary A is given oracle access to O_b (quantum) and to **Reprogram**, and outputs a bit b' .

The distinguishing advantage is

$$|\Pr[\mathbf{Repro}_1^A \implies 1] - \Pr[\mathbf{Repro}_0^A \implies 1]|.$$

Let p_X be the marginal of p on X . Define

$$p_{\max}^{(r)} := \mathbb{E} \left[\max_x p_X^{(r)}(x) \right],$$

where $p^{(r)}$ is the adversary's r -th input to the **Reprogram** oracle, and the expectation is over the adversary's randomness up to round r .

² Without loss of generality, this unitary does not depend on i : We can include a counter register Z_N in Z which keeps track of the number of queries that have been applied thusfar, and effectively apply a unitary U_i after the i -th query by defining U as a controlled unitary with control register Z_N

Lemma 1. [Simplified from Theorem 1 in [GHHM21]] For any adversary D making R reprogramming calls and q quantum queries in total,

$$|\Pr[\text{Repro}_1^D \Rightarrow 1] - \Pr[\text{Repro}_0^D \Rightarrow 1]| \leq \sum_{r=1}^R \left(\sqrt{q \cdot p_{\max}^{(r)}} + \frac{1}{2} q \cdot p_{\max}^{(r)} \right). \quad (2)$$

2.3 Quantum union bound

We need a lemma referred to as the quantum union bound.

Lemma 2 (Theorem 1.1 in [OV22]). Let $|\psi\rangle \in \mathcal{H}$ be a quantum state and let P_i , $i = 1, \dots, N$ be projectors on $\mathcal{H}$ such that $\langle \psi | P_i | \psi \rangle \geq 1 - \epsilon_i$. Then

$$\left\| \left(\prod_{i=1}^N P_i \right) |\psi\rangle \right\|^2 \geq 1 - 4 \sum_{i=1}^N \epsilon_i.$$

2.4 Chernoff Bounds

The following lemma is taken from combining Theorem 4.4 and 4.5 from [MU05].

Lemma 3 (Chernoff Bounds). Let $X_1, \dots, X_n$ be independent Bernoulli variables such that $\Pr(X_i = 1) = p_i$. Let

$$X = \sum_{i=1}^n X_i \quad \text{and} \quad \mu = \mathbb{E}[X].$$

Then the following Chernoff bounds hold:

1. (**Upper tail**) For $0 < \delta \leq 1$,

$$\Pr(X \geq (1 + \delta)\mu) \leq e^{-\mu\delta^2/3}. \quad (4.2)$$

2. (**Lower tail**) For $0 < \delta \leq 1$,

$$\Pr(X \leq (1 - \delta)\mu) \leq e^{-\mu\delta^2/2}. \quad (4.4)$$

The random variables X_i defined by sequential measurements form a martingale difference sequence rather than independent trials. We therefore need to apply the following martingale Chernoff bound derived from Azuma-Hoeffding Inequality below, from Theorem 12.6 in [MU05]

Lemma 4 (Azuma-Hoeffding Inequality). Let $(Y_0, Y_1, \dots, Y_n)$ be a martingale such that for all $i = 1, \dots, n$,

$$|Y_i - Y_{i-1}| \leq c_i \quad \text{almost surely.}$$

Then for any $t > 0$,

$$\Pr[Y_n - Y_0 \geq t] \leq \exp\left(-\frac{t^2}{2 \sum_{i=1}^n c_i^2}\right).$$

3 Proof of Post Quantum Security of the Fischlin Transform

We are now set to prove that the Fischlin transform is a NIZK (Definition 7) with straight-line extractability (Definition 8) and zero knowledge in Quantum-Random Oracle Model.

3.1 Proof Intuition: From Classical Counting to Quantum Extraction

We begin by providing some additional detail about the proof strategy to guide the reader.

We start with the strategy for the deterministic prover. Upper-bounding the probability of getting η outcomes 0^l when applying a hypothetical measurement (used for analysis) to the part of the superposition oracle register corresponding to valid transcripts starting with a fixed commitment vector is straightforward (Lemma 5). The difficult part is lower-bounding that same probability after applying two projectors sequentially, one that ensures the success of the prover, and one that ensures the failure of the extractor. The prover success projector leaves the oracle registers corresponding to the inputs queried by verification in the state $|0^l\rangle$. Changing to the compressed basis leaves most of the k registers in the state $|0^l\rangle$ (in superposition, see Lemma 6). Applying the extractor failure projector now correlates the part of the database register that is used for verifying the proof (and that still contains many 0^l 's) and the remaining database register. The extractor-fail projector ensures, however, that in superposition, most registers are in the state $|\perp\rangle$ (the compressed basis initial state). Switching back to the uncompressed basis now (i) transforms the $|\perp\rangle$ states into uniform superpositions, and again (ii) leaves most of the remaining $|0^l\rangle$'s intact (via another application of Lemma 5, see Eq. (21) in the proof of Lemma 11). Measuring in the computational basis now should yield many 0^l 's from the registers used to verify the proof, and also many 0^l 's as the remaining registers have been projected by the extractor fail projector onto a coarse approximation of the uniform superposition. Showing the latter requires us to symmetrize the state (see Eq. (13)). This facilitates a lower bound on the probability of obtaining outcome 0^l when measuring the i 'th register *conditioned* on the measurement results of the first $i - 1$ registers (Lemma 7) and consequently on the cumulative mean of our super-martingale of measurement results (Lemma 9). This lower bound eventually allows the application of martingale techniques (Lemma 8).

The most involved part of the proof is now done. We lift the result to general malicious provers in Section 3.3. The proof of the main lifting theorem, Theorem 2, involves decomposing a prover's final state according to when it has first queried an input starting with a fixed commitment vector. As the relevant part of the compressed oracle database is untouched before that query, we can *post-select* on that query with input starting with a fixed commitment vector actually being made, without query cost. This allows the application of the result for deterministic provers.

Prover and Extractor Definitions We first describe how the prover and extractor work in our analysis for QROM-extractability of the Fischlin Transform described in Definition 9. While it is obviously important for the security of the NIZK that not all transcripts are valid, we will only explicitly keep track of the part of the compressed oracle register that corresponds to transcripts that are accepted by the Σ -protocol-verifier, as this turns out to be sufficient for extractability.

Definition 10 (Extractor Ext_{Fis}). *Let Σ be a Σ -protocol. The quantum straight-line extractor for the Fischlin transform is given the statement x and simulates the quantum random oracle using a compressed oracle. After the adversary has finished, it receives the proof π output by the adversary.*

The extractor works as follows.

$\text{Ext}_{\text{Fis}}(x) :$

1. *The prover receives oracle access to a compressed oracle with database D and produces a proof π .*
2. *Verify the proof. If verification fails, abort.*
3. *Measure the compressed oracle database register. Let $((\mathbf{a}, i, c, z), (\mathbf{a}, i, c', z'))$ be the lexicographically first pair of inputs in the measured database such that $(c, z) \neq (c', z')$, and $\mathcal{V}_\Sigma(x, a_i, c, z) = \mathcal{V}_\Sigma(x, a_i, c', z') = 1$. If no such query is found, abort.*
4. *Output $\text{Ext}_{\text{ss}}(x, a_i, c, c', z, z')$.*

3.2 Extactability – Deterministic Commitments

We first prove that the extractor will succeed to extract a witness whenever a prover outputs a valid proof *that starts with a fixed vector of commitments*. We will later lift that result to arbitrary provers (see Section 3.3).

In this section, we consider therefore a modified NIZK $\Pi_{\text{Fis}}^{(\mathbf{a}_0)}$ parameterized by a commitment vector $\mathbf{a}_0$ where the verifier only accepts if the proof is valid *and* the proof starts with the commitment vector $\mathbf{a}_0$. We can thus further restrict our attention to only the compressed oracle registers for inputs of the form $(\mathbf{a}, i, c, z)$ such that $\mathbf{a} = \mathbf{a}_0$, and (a_i, c, z) is accepted by $\mathcal{V}_\Sigma$. As we assume the Σ -protocol to have unique responses, any input is specified among the remaining set of inputs by the pair $(i, c) \in [k] \times \mathcal{C}$.

An arbitrary dishonest prover $\mathcal{P}^*$ for the Fischlin transform is a quantum query algorithm as described in Section 2.1, with the output register O having the right structure to hold a proof $\pi = (\mathbf{a}, \mathbf{c}, \mathbf{z})$. The joint state of the prover and the oracle in the uncompressed basis after the unitary part of a malicious prover has finished but O has not been measured to produce the prover's output is denoted as

$$|\phi\rangle = \sum_{\mathbf{x}, H} 2^{-\frac{tkN}{2}} |\mathbf{x}\rangle_O |\phi^{\mathbf{x}, H}\rangle_Z |H\rangle_D \quad (4)$$

Here, $D = (D_{(i,d)})_{i \in [k], d \in \mathcal{C}}$ and we subsume the malicious prover's query input and output registers as well as the remainder of the compressed oracle database

in the register Z . The marginal of the register D remains uniform (as the query operator is a controlled unitary with control register D).

To bound the probability for the event where the prover succeeds and the extractor fails in the compressed oracle database, the analysis in the following sections is used. Below we provide a table for all our notations used in this section.

To facilitate our statistical arguments we define the following projector.

$$P_\eta = \sum_{\bar{\omega}(H) \geq \eta} |H\rangle\langle H|_D$$

where $\bar{\omega}(H)$ is defined as $\bar{\omega}(H) = |\{x \mid H(x) = 0\}|$.

Applying the projector $(P_\eta)_D$ to the state $|\phi\rangle_{OZD}$ yields a vector of small norm for η significantly bigger than $2^{-l}kN$ by a standard Chernoff bound.

Lemma 5. *Let $\mu = 2^{-l}kN$, $\delta > 0$ and $\eta = (1 + \delta)\mu$. We have the following probability bound*

$$\|P_\eta|\phi\rangle\|^2 \leq \exp\left(-\frac{\delta^2\mu}{3}\right) =: \varepsilon''.$$

Proof. Let $X_{i,c}$, $i \in [k]$, $c \in \mathcal{C}$ be random variables defined as measurement results of the state $|\phi\rangle$ by

$$X_{i,c} = \begin{cases} 1 & \text{if measuring register } D_{i,c} \text{ yields outcome } 0^l \\ 0 & \text{else} \end{cases},$$

Clearly $\|P_\eta|\phi\rangle\|^2 = \sum_{i \in [k], c \in \mathcal{C}} X_{i,c}$. Also, measuring the subregisters of D in the computational basis yields independent uniformly random l -bit strings,³ so $\Pr[X_{i,c} = 1] = 2^{-l}$ independently for all (i, c) . By a standard Chernoff bound, Lemma 3, we thus have

$$\mathbb{P}[(S_n \geq \eta = (1 + \delta)\mu)] \leq \mathbb{P}[|S_n - \mu| \geq \delta\mu] \leq \exp\left(-\frac{\delta^2\mu}{3}\right).$$

□

Thinking classically for a moment, the proof strategy is via contradiction: a malicious prover succeeding *and* the extractor failing with significant probability would imply that $\|P_\eta|\phi\rangle\|^2$ must in fact be large, contradicting the above tail bound. As we are dealing with a malicious quantum prover and with the compressed oracle that is directly manipulated (measured) by the extractor, the quantum analogue of this argument is significantly more involved.

Denote $|\phi'\rangle = P_s|\phi\rangle$, where

$$P_s = \sum_{\mathbf{x}} |\mathbf{x}\rangle\langle \mathbf{x}|_O \otimes |0^l\rangle\langle 0^l|_{D_{\text{hi}(\mathbf{x})}}^{\otimes k} \quad (5)$$

³ essentially by the correctness of the superposition oracle methodology

Is the success projector, i.e., the probability that the malicious prover succeeds is $\|\phi'\|^2$. Here, $D_{\text{hi}(\mathbf{x})}$ is the subregister of D containing the hash inputs needed to verify the proof $\mathbf{x}$, i.e., $\text{hi}(\mathbf{x}) = \{(\mathbf{a}, i, c_i, z_i) \mid i \in [k]\}$.

We now see the state of the oracle in the compressed picture. Writing $\mathbf{x} = (\mathbf{a}, \mathbf{c}, \mathbf{z})$, we apply Comp,

$$\begin{aligned} |\phi''\rangle &= \text{Comp} P_s |\phi\rangle = \sum_{\mathbf{x}, y, z} |\mathbf{x}\rangle_O |\phi^{\mathbf{x}}\rangle_{Z D_{\text{hi}^c}} (\text{Comp} |0^l\rangle^{\otimes k})_{D_{\text{hi}(\mathbf{x})}} \\ &= \sum_{\mathbf{x}} |\mathbf{x}\rangle_O |\phi^{\mathbf{x}}\rangle_{Z D_{\text{hi}^c}} \left(\sum_{\substack{S \subset \text{hi}(\mathbf{x}) \\ s=|S| \geq (1-\gamma)k}} |0^l\rangle_{D_S}^{\otimes s} |\Gamma\rangle_{D_{\text{hi}(\mathbf{x}) \setminus S}} + |\delta_{\gamma, k}\rangle_{D_{\text{hi}(\mathbf{x})}} \right), \end{aligned} \quad (6)$$

where we have used Lemma 6, and $0 < \gamma \leq \frac{1}{2}$ is chosen later.

Lemma 6. *For some sub-normalized states $|\Gamma^{(s)}\rangle$, $0 \leq s \leq k$, we have the decomposition*

$$(\text{Comp} |0^l\rangle)^{\otimes k} = \sum_{\substack{S \subset [k] \\ s=|S| \geq (1-\gamma)k}} |0^l\rangle_{D_S}^{\otimes s} |\Gamma^{(|S|)}\rangle_{D_{S^c}} + |\delta_{\gamma, k}\rangle,$$

where $\langle 0 |_i |\Gamma^{(s)}\rangle_i = 0$ on any i th qubit of $|\Gamma\rangle$, and the tail bound is given by $\| |\delta_{\gamma, k}\rangle \|^2 \leq e^{-(\gamma-2\cdot 2^{-l})k/2} =: \varepsilon_{\gamma, k}$.

Proof. We bound

$$\begin{aligned} (\text{Comp} |0^l\rangle)^{\otimes k} &= ((| \perp \rangle \langle +^l | + | +^l \rangle \langle \perp | + (\mathbb{1} - | +^l \rangle \langle +^l | - | \perp \rangle \langle \perp |) |0^l\rangle)^{\otimes k} \\ &= \left(|0^l\rangle + 2^{-l/2} | \perp \rangle - 2^{-l/2} | +^l \rangle \right)^{\otimes k} \\ &= \left(|0^l\rangle + 2^{-l/2} | \perp \rangle - 2^{-l} \sum_{z \in \{0,1\}^l} |z\rangle \right)^{\otimes k} \\ &= \left((1 - 2^{-l}) |0^l\rangle + 2^{-l/2} | \perp \rangle - 2^{-l} \sum_{\substack{z \in \{0,1\}^l \\ z \neq 0^l}} |z\rangle \right)^{\otimes k} \\ &= \sum_{\substack{S \subset [k] \\ s=|S| \geq (1-\gamma)k}} |0^l\rangle_{D_S}^{\otimes s} |\Gamma^{(s)}\rangle_{D_{S^c}} + \sum_{\substack{S \subset [k] \\ s=|S| < (1-\gamma)k}} |0^l\rangle_{D_S}^{\otimes s} |\Gamma^{(s)}\rangle_{D_{S^c}}, \\ &= \sum_{\substack{S \subset [k] \\ s=|S| \geq (1-\gamma)k}} |0^l\rangle_{D_S}^{\otimes s} |\Gamma^{(s)}\rangle_{D_{S^c}} + |\delta_{\gamma, k}\rangle \end{aligned} \quad (7)$$

where $\langle 0 |_{D_i} |\Gamma^{(|S|)}\rangle_{D_{S^c}} = 0$ for all $S \subset [k]$ and $i \in S^c$.

We now bound $||\delta_{\gamma,k}\rangle||^2$. We can use a standard Chernoff bound for the number of results 0^ℓ we would obtain were we to measure the entire state in the computational basis. The mean is

$$\mu = k(1 - 2^{-l})^2$$

Towards using a Chernoff bound we define δ via $(1 - \delta)\mu = (1 - \gamma)k$, i.e.,

$$\begin{aligned}\delta &= 1 - \frac{(1 - \gamma)k}{\mu} \\ &= 1 - \frac{(1 - \gamma)k}{kp} = 1 - \frac{(1 - \gamma)}{p}\end{aligned}$$

As $||\delta_{\gamma,k}\rangle||^2$ is the probability of obtaining less than $(1 - \gamma)k$ outcomes 0^l , we then obtain

$$\begin{aligned}||\delta_{\gamma,k}\rangle||^2 &\leq e^{-\delta^2\mu/2} = e^{-[1-(1-\gamma)/p]^2 \times kp/2} = e^{-[p-(1-\gamma)]^2 k/2p} \\ &\leq e^{-[(1-2^{-l})^2 - (1-\gamma)]^2 k/2p} \leq e^{-[2^{-2l} - 2 \cdot 2^{-l} + \gamma]k/2p} \leq e^{-(\gamma - 2 \cdot 2^{-l})k/2}\end{aligned}$$

□

Analyzing the extractor. Let the orthogonal operators $\mathcal{E}_f$ and $\mathcal{E}_s = \mathbb{1} - \mathcal{E}_f$ to denote the projections onto the subspace (in the compressed picture) where the extractor Ext_{Fis} (definition 10) of the Fischlin Transform fails and succeeds, respectively. The operator $\mathcal{E}_f$ projects onto the states with only one register $D_{\mathbf{a},i,c,z}$ for each i that is not in the state $|\perp\rangle$:

$$(\mathcal{E}_f)_D = \left(\sum_{H: \text{Ext}_{\text{Fis}} \text{ fails}} |H\rangle\langle H| \right)_D = \bigotimes_{i=1}^k (P_{\perp,1}^N)_{D_{i,*}},$$

where $P_{\perp,1}^N$ is the projector onto the subspace

$$W_{1,\perp}^N = \text{span}\{|\psi\rangle_{D_i} |\perp^{N-1}\rangle_{D_{i^c}} \mid |\psi\rangle \in \mathbb{C}^{2^l}\},$$

and for each repetition index $i \in [k]$, we write

$$D_{i,*} := \{D_{(\mathbf{a}_0, i, c, z)} : c \in \mathcal{C}\}$$

for the collection of compressed oracle registers corresponding to all valid transcripts with fixed commitment vector $\mathbf{a}_0$ and repetition index i , ranging over all challenges $c \in \mathcal{C}$.

We need to bound the probability that the extractor fails and the prover succeeds, so we study the norm of the state

$$\begin{aligned}
& \text{Comp}_D \mathcal{E}_f \text{Comp}_D P_s |\phi\rangle \\
&= \text{Comp}_D \mathcal{E}_f \sum_{\mathbf{x}} |\mathbf{x}\rangle_O |\phi^{\mathbf{x}}\rangle_{ZD_{\text{hi}(\mathbf{x})}C} \left(\sum_{\substack{S \subset \text{hi}(\mathbf{x}) \\ s=|S| \geq (1-\gamma)k}} |0^l\rangle_{D_S}^{\otimes s} |\Gamma^{(s)}\rangle_{D_{\text{hi}(\mathbf{x}) \setminus S}} + |\delta_{\gamma,k}\rangle_{D_{\text{hi}(\mathbf{x})}} \right) \\
&= \sum_{\mathbf{x}} |\mathbf{x}\rangle_O \sum_{\substack{S \subset \text{hi}(\mathbf{x}) \\ s=|S| \geq (1-\gamma)k}} |\psi_S^{\mathbf{x}}\rangle_{ZD_{SC}} (\text{Comp}|0^l\rangle_{D_S}^{\otimes s} + |\delta'_{\gamma,k}\rangle_{OZD}) \tag{8}
\end{aligned}$$

where we implicitly define $|\hat{\psi}_S^{\mathbf{x}}\rangle$ by

$$|\hat{\psi}_S^{\mathbf{x}}\rangle_{ZD_{SC}} |0^l\rangle_{D_S}^{\otimes s} = \mathcal{E}_f |\phi^{\mathbf{x}}\rangle_{ZD_{\text{hi}(x)}C} |\Gamma^{(s)}\rangle_{D_{\text{hi}(x) \setminus S}} |0^l\rangle_{D_S}^{\otimes s}$$

and have set $|\psi_S^{\mathbf{x}}\rangle_{ZD_{SC}} = \text{Comp}_{D_{SC}} |\hat{\psi}_S^{\mathbf{x}}\rangle_{ZD_{SC}}$ and

$$|\delta'_{\gamma,k}\rangle_{OZD} = \text{Comp}_D \mathcal{E}_f \sum_{\mathbf{x}} |\mathbf{x}\rangle_O |\phi^{\mathbf{x}}\rangle_{ZD_{\text{hi}(x)}C} (|\delta_{\gamma,k}\rangle_{D_{\text{hi}(\mathbf{x})}}). \tag{9}$$

Note that

$$\| |\delta'_{\gamma,k}\rangle \|^2 \leq \varepsilon_{\gamma,k} \tag{10}$$

since both $\mathcal{E}_f$ and Comp_D are contractions, so the bound follows directly from Lemma 6.

We will now show that measuring a normalized version of

$$|\psi^{\mathbf{x}}\rangle_{ZD} = \sum_{\substack{S \subset \text{hi}(\mathbf{x}) \\ s=|S| \geq (1-\gamma)k}} |\psi_S^{\mathbf{x}}\rangle_{ZD_{SC}} (\text{Comp}|0^l\rangle_{D_S}^{\otimes s}) \tag{11}$$

in the computational basis yields many 0s. Comparing to the final prover-oracle state without conditioning on success and extractor failure yields a bound on $\|\mathcal{E}_f \text{Comp} P_s |\phi\rangle\|^2$, i.e., the probability that the prover succeeds, but the extractor still fails.

For $m, n \in \mathbb{N}$, $m \geq n$, we define the subspaces

$$\begin{aligned}
W_n^m &= \text{span} \left\{ |+\rangle_{D_S}^{\otimes s} |\psi\rangle_{D_{Sc}} \right. \\
&\quad \left. \left| S \subset [m], s := |S| \geq n, |\psi\rangle \in \left(\mathbb{C}^{2^l} \right)^{\otimes m-s} \right\} \subseteq \left(\mathbb{C}^{2^l} \right)^{\otimes m}.
\end{aligned}$$

The projector $\mathcal{E}_f$ projects onto a subspace where for each i there is only one register $D_{\mathbf{a},i,c,z}$ that is not in the state $|\perp\rangle$. As $\text{Comp} |\perp\rangle = |+\rangle$ and $1 \leq i \leq k$, we have

$$|\psi^{\mathbf{x}}\rangle \in W_n^m \otimes \left(\mathbb{C}^{2^l} \right)^k \otimes \mathcal{H}_{XYZ}, \tag{12}$$

where $m = k(N-1)$, $n = m - \gamma k$, W_n^m is viewed as a subspace of the state space of $D_{\text{hi}(\mathbf{x})^c}$ and $\mathcal{H}_{XYZ}$ is the state space of registers XYZ . In the following, we denote $\tilde{D} = D_{\text{hi}(x)^c}$ and its subregisters by $\tilde{D}_i$, $1 \leq i \leq m$. For simplicity, we sometimes write $|\alpha\rangle_I := |\alpha\rangle_{\tilde{D}_I}$ to indicate that for some $I \subset [m]$, the register $\tilde{D}_I$ is in state $|\alpha\rangle$.

We want to bound $\|(P_\eta)_{\tilde{D}} |\psi^{\mathbf{x}}\rangle\|$ for some integer η . This quantity is invariant under permutation of the m subregisters of $\tilde{D}$. We therefore analyze a symmetrized version of the state.

Consider the permutation action of S_m on the subregisters of $\tilde{D}$,

$$\pi |y_1\rangle |y_2\rangle \dots |y_m\rangle = |y_{\pi^{-1}(1)}\rangle |y_{\pi^{-1}(2)}\rangle \dots |y_{\pi^{-1}(m)}\rangle.$$

We can now define the state

$$|\psi_{\text{sym}}^{\mathbf{x}}\rangle_{XZDG} = \frac{1}{\sqrt{m!}} \sum_{\pi \in S_m} \pi_{\tilde{D}} |\psi^{\mathbf{x}}\rangle_{XYZD} |\pi\rangle_G \quad (13)$$

and observe that $\|(P_\eta)_{\tilde{D}} |\psi^{\mathbf{x}}\rangle\| = \|(P_\eta)_{\tilde{D}} |\psi_{\text{sym}}^{\mathbf{x}}\rangle_{XZDG}\|$, since P_η commutes with permutations of the m subregisters. We will now lower-bound $\|(P_\eta)_{\tilde{D}} |\psi_{\text{sym}}^{\mathbf{x}}\rangle_{XZDG}\|$ using the following strategy.

1. We define a sequence of binary random variables indicating whether measuring the $\tilde{D}_i$ resulted in 0^l or not
2. We show a lower bound on the conditional expectations of this sequence of random variables
3. We use the Azuma-Hoeffding martingale concentration inequality to show a lower bound on the number of 0^l outcomes that holds with high probability

We begin with a lemma that will facilitate the lower bound on the conditional expectation. Informally, this lemma shows that, for a symmetric state $|\psi\rangle$ that is supported on subspace W_n^m on register $\tilde{D}$ which is spanned by states with most sub-registers in state $|+^l\rangle$, the probability of obtaining 0^l when measuring the first sub-register is not too small. This is because due to the symmetry assumption, the amplitude of the first sub-register being in state $|+^l\rangle$ is at least $\sqrt{m/n}$.

Lemma 7. *Let $|\psi\rangle_{\tilde{D}E} \in W_n^m \otimes \mathcal{H}_E$ be a normalized quantum state on registers $\tilde{D} = \tilde{D}_1 \dots \tilde{D}_m$ and an additional register E such that for all operators A acting on $\tilde{D}$ and for all $\pi \in S_m$ it holds that $\|A_{\tilde{D}} \pi_{\tilde{D}} |\psi\rangle_{\tilde{D}E}\| = \|A_{\tilde{D}} |\psi\rangle_{\tilde{D}E}\|$. Then*

$$\|\langle 0^l |_{\tilde{D}_1} |\psi\rangle_{\tilde{D}E}\|^2 \geq \frac{2^{-l}n}{m} - 2^{1-l/2} \sqrt{\frac{n(m-n)}{m^2}}$$

Proof. We can decompose

$$|\psi\rangle_{\tilde{D}E} =: \sum_{\substack{I \subset [m] \\ |I| \geq n}} |+^l\rangle_I |\psi_I\rangle_{I^c E}$$

with the convention that $\langle +^l | \psi_I \rangle_{I^c E} = 0$. We can use this decomposition to derive expressions for the parts of the state where the first subregister is in state $|+^l\rangle$, and orthogonal to it, respectively,

$$\begin{aligned}
|\psi\rangle_{\tilde{D}E} &= \sum_{\substack{I \subset [m] \\ |I| \geq n}} |+\rangle_I^{\otimes |I|} |\psi_I\rangle_{I^c E} = \sum_{\substack{I \subset [m] \\ |I| \geq n \\ 1 \in I}} |+\rangle_I^{\otimes |I|} |\psi_I\rangle_{I^c E} + \sum_{\substack{I \subset [m] \\ |I| \geq n \\ 1 \notin I}} |+\rangle_I^{\otimes |I|} |\psi_I\rangle_{I^c E} \\
&= |+\rangle_1 \left(\sum_{\substack{I' \subset [m] \setminus 1 \\ |I'| \geq n-1}} |+\rangle_{I'}^{\otimes |I'|} |\psi_{I' \cup \{1\}}\rangle_{(I' \cup \{1\})^c E} \right) + \sum_{\substack{I \subset [m] \\ |I| \geq n \\ 1 \notin I}} |+\rangle_I^{\otimes |I|} |\psi_I\rangle_{I^c E} \\
&=: |+\rangle_1 |\psi^+\rangle_{1^c E} + |\psi^-\rangle_{\tilde{D}E},
\end{aligned} \tag{14}$$

where $\langle +^l | \psi^-\rangle_{\tilde{D}E} = 0$. By the assumed permutation symmetry of $|\psi\rangle$, the norm of $|\psi_I\rangle$ only depends on $|I|$. We thus define $\zeta_i = \|\psi_{[i]}\|^2$ and note that by the normalization of $|\psi\rangle$ we have

$$\sum_{i=n}^m \binom{m}{i} \zeta_i = 1.$$

We can now bound

$$\begin{aligned}
\| |\psi_+\rangle \|^2 &= \sum_{\substack{I' \subset [m] \setminus 1 \\ |I'| \geq n-1}} \| |\psi_{I' \cup \{1\}}\rangle \|^2 \\
&= \sum_{i=n}^m \binom{m-1}{i-1} \zeta_i \\
&= \sum_{i=n}^m \frac{i}{m} \binom{m}{i} \zeta_i \\
&\geq \sum_{i=n}^m \frac{n}{m} \binom{m}{i} \zeta_i \\
&= \frac{n}{m} =: p
\end{aligned}$$

We can now bound for some $\alpha \in [0, 1]$,

$$\begin{aligned}
\| |0^l\rangle_1 |\psi\rangle_{\tilde{D}E} \|^2 &= \| \langle 0 | + \rangle |\psi^+\rangle_{1^c E} + \langle 0 |_1 |\psi^-\rangle_{\tilde{D}E} \|^2 \\
&\geq \left(2^{-l/2} \| |\psi^+\rangle \| - \alpha \| |\psi^-\rangle \| \right)^2 \\
&\geq \left(2^{-l/2} \sqrt{p} - \alpha \sqrt{1-p} \right)^2 = 2^{-l} p + \alpha^2 (1-p) - \alpha 2^{1-l/2} \sqrt{p(1-p)} \\
&\geq \frac{2^{-l} n}{m} - 2^{1-l/2} \sqrt{\frac{n(m-n)}{m^2}}.
\end{aligned}$$

□

Bounding $\|(P_\eta)_{\tilde{D}}|\psi^{\mathbf{x}}\rangle\|$ using martingale tail bound. We will now follow the same strategy as in Lemma 5 to show that after applying the extractor failure projector, there are, in fact, more than η registers in state $|0^l\rangle$ with high probability, with one crucial difference: the random variables obtained by measuring the sub-registers of $\tilde{D}$ in the state $|\psi^{\mathbf{x}}\rangle$ are not independent. This renders the standard Chernoff bound we used in Lemma 5 inapplicable. We therefore resort to a martingale tail bound for our analysis.

In the following, for ease of notation, let

$$|\psi\rangle = \frac{1}{\|\psi_{\text{sym}}^{\mathbf{x}}\|} |\psi_{\text{sym}}^{\mathbf{x}}\rangle.$$

Let X_i , $0 \leq i \leq k(N-1)$ be the random variables defined analogously to the proof of Lemma 5, i.e.

$$X_i = \begin{cases} 1 & \text{if measuring register } \tilde{D}_i \text{ of state } |\psi\rangle \text{ yields } 0^l \\ 0 & \text{else.} \end{cases}$$

We define $X_{<i} = (X_1, \dots, X_{i-1})$, and

$$Z_i = \mathbb{E}[X_i | X_{<i}], \quad (15)$$

$$\hat{X}_i = Z_i - X_i, \text{ and} \quad (16)$$

$$\hat{Z}_i = \sum_{j=1}^i \hat{X}_j. \quad (17)$$

Note that conditioning on $X_{<i}$, $Z_{<i}$, $\hat{X}_i$ and $\hat{Z}_{<i}$ are all equivalent. Let the value of the standard cumulative mean be denoted as

$$\mu' = \sum_{i=1}^n \mathbb{E}[X_i | X_{<i}] = \sum_{i=1}^n Z_i. \quad (18)$$

Lemma 8. *We have the following probability bound*

$$\|(P_\eta)_{\tilde{D}}|\psi\rangle\|^2 \geq 1 - \exp\left(-\frac{\delta'^2 \underline{\mu}'^2}{2m}\right) =: 1 - \varepsilon'$$

Here,

$$\underline{\mu}' := 2^{-l}k \left[N - 1 - \gamma \left(1 + 4 \cdot 2^l + \log \left(\frac{\log k}{4\gamma} \right) \right) - 4\sqrt{2^l \gamma N} \right]$$

and

$$\eta = (1 - \delta') \underline{\mu}'.$$

Proof. As in the proof of Lemma 5, we use the fact that

$$\|(P_\eta)_{\tilde{D}}|\psi\rangle\|^2 = \Pr \left[\sum_{i=1}^m X_i \geq \eta \right].$$

Note that conditioning on $X_{<i}$, $Z_{<i}$, $\hat{X}_i$ and $\hat{Z}_{<i}$ are all equivalent. We get

$$\begin{aligned}\mathbb{E}[\hat{X}_i \mid \hat{Z}_{<i}] &= \mathbb{E}[Z_i - X_i \mid X_{<i}] \\ &= \mathbb{E}[\mathbb{E}[X_i \mid X_{<i}] - X_i \mid X_{<i}] \\ &= \mathbb{E}[X_i \mid X_{<i}] - \mathbb{E}[X_i \mid X_{<i}] \\ &= 0\end{aligned}$$

$$\implies \mathbb{E}[\hat{Z}_i \mid \hat{Z}_{<i}] = \hat{Z}_{i-1} + \mathbb{E}[\hat{X}_i \mid \hat{Z}_{<i}] = \hat{Z}_{i-1}$$

Hence $\hat{Z}_i$ is a martingale. Additionally, $\hat{Z}_0 = 0$ and $|\hat{Z}_i - \hat{Z}_{i-1}| = |\hat{X}_i| \leq 1$. Applying Lemma 4 we thus get

$$\begin{aligned}\Pr\left[\sum_{j=1}^m X_j \leq \mu' - \epsilon\right] &= \Pr\left[\sum_{j=1}^m X_j \leq \sum_{j=1}^m Z_j - \epsilon\right] \\ &= \Pr[0 - \hat{Z}_m \leq -\epsilon] \\ &\leq \exp\left(-\frac{\epsilon^2}{2m}\right).\end{aligned}$$

Using the lower bound on the cumulative mean from Lemma 9 we obtain the desired bound,

$$\Pr\left[\sum_{j=1}^m X_j \leq \underline{\mu}' - \epsilon\right] \leq \exp\left(-\frac{\epsilon^2}{2m}\right).$$

□

We continue by providing the needed lower bound on the cumulative mean.

Lemma 9. *The cumulative mean defined in Eq. (18) is bounded as*

$$\mu' \geq \underline{\mu}' = 2^{-l}k \left[N - 1 - \gamma \left(1 + 4 \cdot 2^l + \log \left(\frac{4\gamma}{\log k} \right) \right) - 4\sqrt{2^l \gamma N} \right].$$

Proof. Let $|\beta\rangle_{\tilde{D}_{[i+1:m']}}$ be the post-measurement state after the first $i-1$ sub-registers of $\tilde{D}$ have been measured and fixed outcomes $y_1, \dots, y_{i-1}$ have been obtained. Then $|\beta\rangle \in W_{n-i+1}^{m-i+1}$. When measuring the first remaining register of $|\beta\rangle$, the probability of obtaining outcome $y_i = 0^l$ is thus lower-bounded as

$$\Pr[X_i = 1 | y_1, \dots, y_{i-1}] \geq \frac{2^{-l}(n-i+1)}{(m-i+1)} - 2^{1-l/2} \sqrt{\frac{(n-i+1)(m-n)}{(m-i+1)^2}}$$

by Lemma 7 and therefore in particular

$$\begin{aligned}\mathbb{E}[X_i|X_{<i}] &= \Pr[X_i = 1|X_{<i}] \geq \frac{2^{-l}(n-i+1)}{(m-i+1)} - 2^{1-l/2} \sqrt{\frac{(n-i+1)(m-n)}{(m-i+1)^2}} \\ &\geq \frac{2^{-l}(n-i)}{(m-i)} - 2^{1-l/2} \sqrt{\frac{(n-i)(m-n)}{(m-i)^2}}.\end{aligned}$$

We can now lower-bound now for any $n' \leq n$,

$$\begin{aligned}\mu' &= \sum_{i=1}^n \mathbb{E}[X_i|X_{<i}] \\ &\geq \sum_{i=1}^{n'} \mathbb{E}[X_i|X_{<i}] \\ &\geq \sum_{i=1}^{n'} \left(\frac{n-i}{m-i} \right) 2^{-l} - 2^{1-l/2} \sqrt{\frac{(n-i)(m-n)}{(m-i)^2}} \\ &= \sum_{i=1}^{n'} \left(1 - \frac{m-n}{m-i} \right) 2^{-l} - 2^{1-l/2} \sqrt{\frac{(n-i)(m-n)}{(m-i)^2}} \\ &= n' - 2^{-l}(m-n) \sum_{i=1}^{n'} \frac{1}{m-i} - \sum_{i=1}^{n'} 2^{1-l/2} \sqrt{\frac{(n-i)(m-n)}{(m-i)^2}}.\end{aligned}$$

We have the following inequalities for $n' < m$ on the above terms in the sum:

$$\begin{aligned}1. \text{ On the second term: } &\sum_{i=1}^{n'} \frac{1}{m-i} \leq \int_1^{n'} \frac{dx}{m-x} = \log \left(\frac{m-1}{m-n'} \right), \\ 2. \text{ On the third term: } &\sum_{i=1}^{n'} \sqrt{\frac{(n-i)}{(m-i)^2}} \leq \sum_{i=1}^{n'} \sqrt{\frac{(m-i)}{(m-i)^2}} \\ &\leq \int_1^{n'} \frac{dx}{\sqrt{m-x}} = 2 \left[\sqrt{(m-1)} - \sqrt{(m-n')} \right]\end{aligned}$$

Hence we get:

$$\begin{aligned}
\mu' &\geq 2^{-l} \left[n' - (m - n) \log \left(\frac{m-1}{m-n'} \right) \right] - 2^{2-l/2} \sqrt{m-n} \left[\sqrt{(m-1)} - \sqrt{(m-n')} \right] \\
&\geq 2^{-l} \left[n' + (m - n) \log \left(\frac{m-n'}{m-1} \right) \right] - 2^{2-l/2} \sqrt{m(m-n)} \\
&= 2^{-l} \left[k(N-1-\gamma) - 2^{2+l} k \gamma + (k \gamma) \log \left(\frac{\gamma(1+2^{2+l})}{2^l \log k - 1} \right) \right] - 2^{2-l/2} \sqrt{m(m-n)} \\
&\geq 2^{-l} k \left[(N - (1 + \gamma + 2^{2+l} \gamma) + (\gamma) \log \left(\frac{4\gamma}{\log k} \right)) \right] - 2^{2-l/2} \sqrt{k(N-1)\gamma k} \\
&\geq 2^{-l} k \left[(N - 1 - \gamma \left(1 + 2^{2+l} + \log \left(\frac{4\gamma}{\log k} \right) \right)) - 2^{2+l/2} \sqrt{N\gamma} \right]
\end{aligned}$$

Here, we have chose n' so that we do not unnecessarily lower-bound non-negative terms by negative ones,

$$\begin{aligned}
&\left\lfloor \frac{n-i}{m-i} \right\rfloor 2^{-l} - \frac{2^{1-l} \sqrt{(n-i)(m-n)}}{m-i} \geq 0 \\
&\implies (n-i) 2^{-l} \geq 2^{1-l/2} \sqrt{(n-i)(m-n)} \\
&\implies (n-i)^2 2^{-2l} \geq 2^{2-l} (n-i)(m-n) \\
&\implies (n-i) \geq 2^{2+l} (m-n) \\
&\implies i \leq n - 2^{2+l} (m-n) =: n'.
\end{aligned} \tag{19}$$

□

We remark that the parameter restrictions of Lemma 10 imply $n' \geq 0$, so the truncation in the proof is non-empty.

QROM Extractability Bound We are now finally ready to prove our main QROM extractability result for deterministic commitments. For brevity, define $\tilde{\mathcal{E}}_f = \text{Comp} \mathcal{E}_f \text{Comp}$ and $\tilde{\mathcal{E}}_s = \text{Comp} \mathcal{E}_s \text{Comp}$. We make the convention that any projectors act on register D unless specified otherwise.

Lemma 10 (Extractability – Deterministic Commitments). *Let $\mathcal{P}$ be a dishonest prover for the Fischlin transform of a special-sound Σ -protocol for a relation R with unique responses. For some constant c , let the challenge space be of size $N = c \cdot 2^l \cdot \log(k)$. Let further $2^{\frac{1}{c}} \leq k \leq 2^{\frac{2^l}{256 \cdot c}}$ and $l \geq 14$. The probability of the event where the prover succeeds and the extractor fails is bounded by*

$$\mathbb{P}[\text{Extractor fails and Prover Succeeds}] = \|\tilde{\mathcal{E}}_f P_s |\phi\rangle\|^2 \leq \varepsilon / (1 - \varepsilon) = \text{negl}(k),$$

$$\text{where } \varepsilon \leq 3 \exp \left[-\frac{k}{128 \cdot c \cdot 2^l \cdot \log k} \right] + 6 \exp \left[-\frac{k}{8 \cdot 2^l} \right].$$

Note that the parameter restrictions allow picking $l = \Theta(\log \log k)$ which indeed makes the error negligible in k .

Proof. Using Lemma 5, we bound

$$\begin{aligned}
\varepsilon'' &\geq \|P_\eta |\phi\rangle\|^2 \\
&\geq \|P_s P_\eta |\phi\rangle\|^2 = \|P_\eta P_s |\phi\rangle\|^2 \\
&= \|P_\eta (\tilde{\mathcal{E}}_s + \tilde{\mathcal{E}}_f) |\phi'\rangle\|^2 \\
&\geq \|P_\eta \tilde{\mathcal{E}}_s |\phi'\rangle\|^2 + \|P_\eta \tilde{\mathcal{E}}_f |\phi'\rangle\|^2 - 2 \operatorname{Re} \langle \phi' | \tilde{\mathcal{E}}_s P_\eta \tilde{\mathcal{E}}_f | \phi' \rangle \\
&\geq \|P_\eta \tilde{\mathcal{E}}_f |\phi'\rangle\|^2 - 2 \operatorname{Re} \langle \phi' | \tilde{\mathcal{E}}_s P_\eta \tilde{\mathcal{E}}_f | \phi' \rangle \\
&\geq (1 - \varepsilon') \|\tilde{\mathcal{E}}_f |\phi'\rangle\|^2 - 4\sqrt{\varepsilon_{\gamma, (1-\gamma)k}} - 2 \operatorname{Re} \langle \phi' | \tilde{\mathcal{E}}_s P_\eta \tilde{\mathcal{E}}_f | \phi' \rangle \\
&\quad \text{(here } \exp\left[-\frac{\delta'^2 \mu'}{2}\right] = \varepsilon' \text{ as in lemma 11, applying lemma 11)} \\
&\geq (1 - \varepsilon') \|\tilde{\mathcal{E}}_f |\phi'\rangle\|^2 - 4\sqrt{\varepsilon_{\gamma, (1-\gamma)k}} - \sqrt{2\varepsilon' + 8\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}} \quad \text{(due to (20) below)}
\end{aligned}$$

In the second line, we have used that P_η and P_s are both diagonal in the standard basis on D . The bound on the inner product expression is obtained as

$$\begin{aligned}
\operatorname{Re} \langle \phi' | \tilde{\mathcal{E}}_s P_\eta \tilde{\mathcal{E}}_f | \phi' \rangle &\leq \langle \phi' | (\mathbb{I} - \tilde{\mathcal{E}}_f) \tilde{\mathcal{E}}_f | \phi' \rangle + \langle \phi' | (\mathbb{I} - \tilde{\mathcal{E}}_f) (P_\eta \tilde{\mathcal{E}}_f - \tilde{\mathcal{E}}_f) | \phi' \rangle \\
&\leq \|(\mathbb{I} - \tilde{\mathcal{E}}_f)\| \| (P_\eta \tilde{\mathcal{E}}_f - \tilde{\mathcal{E}}_f) | \phi' \rangle \| \leq \sqrt{2\varepsilon' + 8\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}} \quad (20) \\
&\quad \text{[Due to Lemma 12].}
\end{aligned}$$

$$\begin{aligned}
&\implies \mathbb{P}[\text{Extractor fails and Prover Succeeds}] = \|\tilde{\mathcal{E}}_f P_s |\phi\rangle\|^2 \\
&\leq \frac{\varepsilon'' + \sqrt{2\varepsilon' + 8\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}} + 4\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}}{1 - \varepsilon'} \leq \frac{\varepsilon}{1 - \varepsilon}
\end{aligned}$$

$$\begin{aligned}
\text{where } \varepsilon &= \max\{\varepsilon', \varepsilon'' + \sqrt{2\varepsilon' + 8\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}} + 4\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}\} \\
&= \varepsilon'' + \sqrt{2\varepsilon' + 8\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}} + 4\sqrt{\varepsilon_{\gamma, (1-\gamma)k}} \\
&\leq \varepsilon'' + \sqrt{2\varepsilon'} + \sqrt{8\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}} + 4\sqrt{\varepsilon_{\gamma, (1-\gamma)k}} \quad (\text{as } \sqrt{a^2 + b^2} \leq |a| + |b|) \\
&\leq \varepsilon'' + 2\sqrt{\varepsilon'} + 7\sqrt{\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}} \\
&\leq 3 \exp\left[-\frac{k}{128 \cdot c \cdot 2^l \cdot \log k}\right] + 7 \exp\left[-\frac{k}{8 \cdot 2^l}\right].
\end{aligned}$$

The last inequality is obtained by crudely balancing errors, which is done in Lemma 13 in the Appendix. $\square$

Lemma 11.

$$\|P_\eta \tilde{\mathcal{E}}_f |\phi'\rangle\|^2 \geq (1 - \varepsilon') \|\tilde{\mathcal{E}}_f |\phi'\rangle\|^2 - 4\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}$$

where $\varepsilon_{\gamma, (1-\gamma)k}$ is tail bound as in lemma 6, $\underline{\mu}'$ is as in Lemma 8,

$$\eta - (1 - 2\gamma)k = (1 - \delta')\underline{\mu}',$$

and

$$\varepsilon' = \exp\left(-\frac{\delta'^2 \underline{\mu}'^2}{2m}\right)$$

Proof. Consider $\tilde{\mathcal{E}}_f P_s |\phi\rangle = \tilde{\mathcal{E}}_f |\phi'\rangle$ and

$$\begin{aligned} |\tilde{\phi}'\rangle &= \frac{\tilde{\mathcal{E}}_f |\phi'\rangle}{\|\tilde{\mathcal{E}}_f |\phi'\rangle\|} \\ &= \sqrt{\Gamma} \sum_{\mathbf{x}} \sqrt{p_{\mathbf{x}}} |\mathbf{x}\rangle_O |\psi_{\text{norm}}^{\mathbf{x}}\rangle_{ED} + \frac{|\delta'_{\gamma,k}\rangle}{\|\tilde{\mathcal{E}}_f |\phi'\rangle\|}. \end{aligned}$$

Here we have used Eqs. (8), (9) and (11) and defined

$$\begin{aligned} |\psi_{\text{norm}}^{\mathbf{x}}\rangle &= \frac{|\psi^{\mathbf{x}}\rangle}{\| |\psi^{\mathbf{x}}\rangle \|}, \text{ the probability distribution} \\ p_{\mathbf{x}} &= \frac{\| |\psi^{\mathbf{x}}\rangle \|^2}{\sum_{\mathbf{x}} \mathbf{x} \| |\psi^{\mathbf{x}}\rangle \|^2} \text{ and} \\ \Gamma &= \frac{\sum_{\mathbf{x}} \mathbf{x} \| |\psi^{\mathbf{x}}\rangle \|^2}{\|\tilde{\mathcal{E}}_f |\phi'\rangle\|^2} \end{aligned}$$

We bound

$$\begin{aligned} \|P_{\eta} |\tilde{\phi}'\rangle\|^2 &\geq \Gamma \mathbb{E}_{\mathbf{x} \leftarrow p} [\|P_{\eta} |\psi_{\text{norm}}^{\mathbf{x}}\rangle\|^2] - 2\sqrt{\epsilon_{\gamma,k}} \\ &= \Gamma \mathbb{E}_{\mathbf{x} \leftarrow p} \left[\left\| P_{\eta} \frac{1}{\| |\psi_{\text{sym}}^{\mathbf{x}}\rangle \|} |\psi_{\text{sym}}^{\mathbf{x}}\rangle \right\|^2 \right] - 2\sqrt{\epsilon_{\gamma,k}} \end{aligned}$$

where we have used Eq. (10). Again using the simplifying notation $|\psi\rangle = \frac{1}{\| |\psi_{\text{sym}}^{\mathbf{x}}\rangle \|} |\psi_{\text{sym}}^{\mathbf{x}}\rangle$ we bound

$$\|P_{\eta} |\psi\rangle_{ED}\|^2 \geq \| (P_{(1-2\gamma)k})_{D_{\text{hi}(\mathbf{x})}} (P_{\eta-(1-2\gamma)k})_{D_{\text{hi}(\mathbf{x})}^c} |\psi\rangle_{ED} \|^2$$

as choosing a particular distribution of the $\eta |0^l\rangle$'s makes the norm smaller. Defining

$$|\psi'\rangle = (P_{\eta-(1-2\gamma)k})_{D_{\text{hi}(\mathbf{x})}^c} |\psi\rangle_{ED}$$

we have that

$$\| |\psi'\rangle \|^2 \geq 1 - \epsilon'$$

due to Lemma 8 (and ϵ' is defined in that same lemma), and

$$|\psi'\rangle = \sum_{\substack{S \subset \text{hi}(\mathbf{x}) \\ s=|S| \geq (1-\gamma)k}} \left| \tilde{\psi}_S^{\mathbf{x}} \right\rangle_{ZD_{SC}} (\text{Comp}|0^l\rangle)_{D_S}^{\otimes s}$$

for some states $\left| \tilde{\psi}_S^{\mathbf{x}} \right\rangle_{ZD_{SC}}$ due to Eq. (8). Applying Lemma 6 with $(1-\gamma)k$ in place of k we get

$$|\psi'\rangle = \sum_{\substack{S \subset \text{hi}(\mathbf{x}) \\ s=|S| \geq (1-2\gamma)k}} \left| \tilde{\psi}_S^{\mathbf{x}} \right\rangle_{ZD_{SC}} |0^l\rangle_{D_S}^{\otimes s} + |\delta'_{\gamma, (1-\gamma)k}\rangle \quad (21)$$

for some states $\tilde{\psi}^{\mathbf{x}}$, where

$$\| |\delta'_{\gamma, (1-\gamma)k}\rangle \|^2 \leq \varepsilon_{\gamma, (1-\gamma)k}.$$

It follows that

$$\begin{aligned} \| (P_{(1-2\gamma)k})_{D_{\text{hi}(\mathbf{x})}} (P_{\eta-(1-2\gamma)k})_{D_{\text{hi}(\mathbf{x})}^c} |\psi\rangle_{ED} \|^2 &= \| (P_{(1-2\gamma)k})_{D_{\text{hi}(\mathbf{x})}} |\psi'\rangle \|^2 \\ &\geq 1 - \epsilon' - 2\varepsilon_{\gamma, (1-\gamma)k}. \end{aligned}$$

In summary, we have

$$\| P_\eta |\psi\rangle_{ED} \|^2 \geq 1 - \epsilon' - 2\varepsilon_{\gamma, (1-\gamma)k}$$

And thus

$$\begin{aligned} \| P_\eta \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 &\geq \| \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 (1 - \epsilon' - 2\varepsilon_{\gamma, (1-\gamma)k}) - 2\sqrt{\varepsilon_{\gamma, k}} \\ &\geq \| \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 (1 - \epsilon') - 4\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}. \end{aligned}$$

□

Lemma 12. $\| (P_\eta \tilde{\mathcal{E}}_f - \tilde{\mathcal{E}}_f) |\phi'\rangle \|^2 \leq 2\varepsilon' + 8\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}.$

Proof. Directly expanding:

$$\begin{aligned} \| (P_\eta \tilde{\mathcal{E}}_f - \tilde{\mathcal{E}}_f) |\phi'\rangle \|^2 &= \| P_\eta \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 + \| \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 - 2\langle \phi' | \tilde{\mathcal{E}}_f P_\eta \tilde{\mathcal{E}}_f | \phi' \rangle \\ &= \| P_\eta \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 + \| \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 - 2\| P_\eta \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 \\ &\leq \| P_\eta \|^2 \| \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 + \| \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 - 2(1 - \varepsilon') \| \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 \\ &\quad + 8\sqrt{\varepsilon_{\gamma, (1-\gamma)k}} (\text{due to lemma 11}) \\ &= (1 + 1 - 2(1 - \varepsilon')) \| \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 + 8\sqrt{\varepsilon_{\gamma, (1-\gamma)k}} \\ &= 2\varepsilon' \| \tilde{\mathcal{E}}_f |\phi'\rangle \|^2 + 8\sqrt{\varepsilon_{\gamma, (1-\gamma)k}} \\ &\leq 2\varepsilon' + 8\sqrt{\varepsilon_{\gamma, (1-\gamma)k}} \end{aligned}$$

□

3.3 Extractability – lifting to general prover.

We now go on to apply our bound for a prover outputting a proof with a fixed value for $\mathbf{a}$ to obtain a bound for a general prover. To this end, we prove a bound on the probability that a general prover succeeds but the extractor fails, in terms of the probability that the same happens for a fixed vector of commitments $\mathbf{a}$.

Theorem 2. *Let $\mathcal{P}$ be a dishonest prover for the Fischlin transform of a special-sound Σ -protocol for a relation R with unique responses, in the QROM, making no more than q quantum queries to the random oracle. We have*

$$\Pr_{(x,\pi,w) \leftarrow \langle \mathcal{P}, \mathcal{E} \rangle} [\mathcal{V}(x, \pi) = 1] = \Pr_{(x,\pi) \leftarrow \mathcal{P}^H} [\mathcal{V}(x, \pi) = 1] \quad (22)$$

and

$$\Pr_{(x,\pi,w) \leftarrow \langle \mathcal{P}, \mathcal{E} \rangle} [\mathcal{V}(x, \pi) = 1 \wedge (x, w) \notin R] \leq 4(q + k)^2 \cdot \delta, \quad (23)$$

where δ is such that for any prover $\mathcal{P}'$ and any $\mathbf{a}_0 \in \mathcal{M}^k$,

$$\Pr_{(x,\pi=(\mathbf{a},\mathbf{c},\mathbf{z}),w) \leftarrow \langle \mathcal{P}', \mathcal{E} \rangle} [\mathbf{a} = \mathbf{a}_0 \wedge \mathcal{V}(x, \pi) = 1 \wedge (x, w) \notin R] \leq \delta. \quad (24)$$

Combining with Lemma 10, we get our main result.

Corollary 1 (straight-line extractability of the Fischlin transform in the QROM). *Let Σ be a Σ -protocol for a relation R with special soundness and unique responses. If the challenge space is of size $N = c \cdot 2^l \cdot \log(k)$ for some constant c and parameters k, l of the Fischlin transform, $2^{\frac{1}{c}} \leq k \leq 2^{\frac{l}{256-c}}$, $l \geq 14$, and $l = O(\log \log k)$ then $\text{Fis}[\Sigma]$ is a proof of knowledge with straight-line extractability. The extractor's simulation of the quantum-accessible random oracle can be made perfect ($\varepsilon_{\text{sim}} = 0$) and for a q -query adversarial prover,*

$$\varepsilon_{\text{ex}} \leq q^2 \cdot \text{negl}(k)$$

A quick note on instantiation. Given a Σ -protocol $\hat{\Sigma}$ with constant challenge space of size $\hat{N}$ we can pick k (not too small) first. Now we can choose c and $l = \lceil \log \log k + \log c + 8 \rceil$, then set $r = \lceil \log_{\hat{N}}(c \cdot 2^l \cdot \log(k)) \rceil$ to allow for the desired challenge space size, and define the Σ -protocol Σ as the r -fold parallel repetition of $\hat{\Sigma}$ with the challenge space artificially restricted to size $c \cdot 2^l \cdot \log(k)$.⁴ The combination of Σ , k, l, c now fulfils the prerequisites of Corollary 1.

Proof (of Theorem 2). The statement in Equation (22) is immediate by the perfect correctness of the compressed oracle.

We begin by constructing a family of provers that either output a valid proof with a fixed commitment vector $\hat{\mathbf{a}}$, or a dummy symbol. Construct prover $\tilde{\mathcal{P}}_{\hat{\mathbf{a}}}$ as follows.

1. Run $(x, \pi = (\mathbf{a}, \mathbf{c}, \mathbf{z})) \leftarrow \mathcal{P}^H$.

⁴ We ignore rounding here, which changes c slightly.

2. Run $b \leftarrow \mathcal{V}^H(x, \pi)$.
3. If $b = 1$ and $\mathbf{a} = \hat{\mathbf{a}}$, output (x, π) , else, output $\perp$.

Let $\mathcal{P}_{\hat{\mathbf{a}}}$ be a version of $\tilde{\mathcal{P}}_{\hat{\mathbf{a}}}$ with all measurements delayed to the end, i.e., where verification is run coherently. Observe that $\mathcal{P}_{\hat{\mathbf{a}}}$ makes at most $q' = q + k$ queries. Without loss of generality, $\mathcal{P}_{\hat{\mathbf{a}}}$ starts with some initial state $|\phi_0\rangle$, applies $(UO^H)^{q'}$ and measures the output register O . We now consider $\mathcal{P}_{\hat{\mathbf{a}}}$ as run by the extractor, i.e., it gets query access to the compressed oracle instead of a standard quantum-accessible random oracle. Denote by $\Xi = |\mathcal{M}|^k \cdot k \cdot N \cdot |\mathcal{Z}|$ the cardinality of the domain of the random oracle. We decompose the final state after the unitary part of $\mathcal{P}_{\hat{\mathbf{a}}}$ as

$$\begin{aligned} & (UO^H)^{q'} |\phi_0\rangle_{XYE} |+\rangle_D^{\otimes \Xi} \\ &= (UO^H(\mathbb{1} - |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|)_{X_{\mathcal{M}}})^{q'} |\phi_0\rangle_{XYE} |+\rangle_D^{\otimes \Xi} \\ &+ \sum_{i=1}^{q'} (UO^H)^{q'-i+1} |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|_{X_{\mathcal{M}}} (UO^H(\mathbb{1} - |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|)_{X_{\mathcal{M}}})^{i-1} |\phi_0\rangle_{XYE} |+\rangle_D^{\otimes \Xi}. \end{aligned}$$

Note that by construction, $\mathcal{P}_{\hat{\mathbf{a}}}$ always queries an input starting with the $\mathbf{a}$ in its output proof (if it outputs one). Therefore we have

$$P_s (UO^H(\mathbb{1} - |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|)_{X_{\mathcal{M}}})^{q'} |\phi_0\rangle_{XYE} |+\rangle_D^{\otimes \Xi} = 0.$$

Now note that by construction,

$$|\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|_{X_{\mathcal{M}}} (UO^H(\mathbb{1} - |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|)_{X_{\mathcal{M}}})^{i-1} |\phi_0\rangle_{XYE} |+\rangle_D^{\otimes \Xi} = |\tilde{\psi}_i\rangle_{XYED_{\hat{\mathbf{a}}}^c} |+\rangle_{D_{\hat{\mathbf{a}}}}^{\otimes ([k] \cdot N \cdot |\mathcal{Z}|)},$$

for some sub-normalized state $|\tilde{\psi}_i\rangle$, as before each query, the input register is projected away from $|\hat{\mathbf{a}}\rangle$.⁵ Let $|\psi_i\rangle = \|\tilde{\psi}_i\rangle\|^{-1} |\tilde{\psi}_i\rangle$. We now define, for each pair $(\hat{\mathbf{a}}, i) \in \mathcal{M}^k \times [q]$, a new prover $\mathcal{P}_{\hat{\mathbf{a}}, i}^H$ as follows.

1. Prepare $|\psi_i\rangle$ as follows:
 - (a) Initialize a compressed oracle for a random function H' identically distributed as H .
 - (b) Run $\mathcal{P}_{\hat{\mathbf{a}}}^{H'}$ until right before the i -th query, but for each query, apply the binary measurement whether the query input starts with $\hat{\mathbf{a}}$. If it does, abort and restart from (a).
 - (c) Finally, apply the same measurement to the query input register. If it returns “not $\hat{\mathbf{a}}$ ”, abort and restart from (a).
2. Run $(x, \pi) \leftarrow \mathcal{P}_{\hat{\mathbf{a}}, i}^{H''}$ from its i -th query, starting with the prepared joint state $|\psi_i\rangle |+\rangle^{\otimes ([k] \cdot N \cdot |\mathcal{Z}|)}$ of prover and H' oracle database, where

$$H''(\mathbf{a}, i, c, z) = \begin{cases} H(\mathbf{a}, i, c, z) & \text{if } \mathbf{a} = \hat{\mathbf{a}} \\ H'(\mathbf{a}, i, c, z) & \text{else.} \end{cases}$$

⁵ Sub-normalization follows from $\sum_i \|\tilde{\psi}_i\rangle\|^2 \leq 1$ which holds by orthogonality of the decomposition.

Here, it is understood that H' is still instantiated with the compressed oracle simulated by the prover $\mathcal{P}_{\hat{\mathbf{a}},i}^H$, while H is the oracle the prover $\mathcal{P}_{\hat{\mathbf{a}},i}^H$ has access to.

3. Output (x, π) .

This prover might not be time-efficient. This is not a problem, however, as all extractability statements hold for provers only (at most) fulfilling a query bound.

We can now relate the probability that $\mathcal{P}$ succeeds and the extractor fails to the provers $\mathcal{P}_{\hat{\mathbf{a}},i}^H$ as follows,

$$\begin{aligned}
& \Pr_{(x,\pi,w) \leftarrow \langle \mathcal{P}, \mathcal{E} \rangle} [\mathcal{V}(x, \pi) = 1 \wedge (x, w) \notin R] \\
&= \sum_{\hat{\mathbf{a}}} \left\| \mathcal{E}_f P_s \sum_{i=1}^{q'} (UO^H)^{q'-i+1} |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|_{X_{\mathcal{M}}} (UO^H (\mathbb{1} - |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|)_{X_{\mathcal{M}}})^{i-1} |\phi_0\rangle_{XYE} |+\rangle_D^{\otimes \Xi} \right\|^2 \\
&\leq \sum_{\hat{\mathbf{a}}} \left(\sum_{i=1}^{q'} \left\| \mathcal{E}_f P_s (UO^H)^{q'-i+1} \right. \right. \\
&\quad \left. \left. |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|_{X_{\mathcal{M}}} (UO^H (\mathbb{1} - |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|)_{X_{\mathcal{M}}})^{i-1} |\phi_0\rangle_{XYE} |+\rangle_D^{\otimes \Xi} \right\| \right)^2 \\
&\leq \delta \sum_{\hat{\mathbf{a}}} \left(\sum_{i=1}^{q'} \left\| |\tilde{\psi}_i\rangle \right\| \right)^2,
\end{aligned}$$

where the first inequality is the triangle inequality and the second inequality results from assumption (24) applied to the provers $\mathcal{P}_{\hat{\mathbf{a}},i}^H$. It remains to bound the sum of norms on the right-hand side. Define

$$\epsilon_{\hat{\mathbf{a}},i} = \left\| |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}| (UO)^{i-1} |\phi_0\rangle |+\rangle_D^{\otimes \Xi} \right\|^2.$$

We can now bound, using Cauchy-Schwarz in the first step,

$$\begin{aligned}
& \frac{1}{q'} \left(\sum_{i=1}^{q'} \left\| |\tilde{\psi}_i\rangle \right\| \right)^2 \leq \sum_{i=1}^{q'} \left\| |\tilde{\psi}_i\rangle \right\|^2 \\
&= \sum_{i=1}^{q'} \langle \phi_0 |_{XYE} \langle + |_D^{\otimes \Xi} \left((\mathbb{1} - |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|)_{X_{\mathcal{M}}} (O^H)^\dagger U^\dagger \right)^{i-1} \\
&\quad |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|_{X_{\mathcal{M}}} (UO^H (\mathbb{1} - |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|)_{X_{\mathcal{M}}})^{i-1} |\phi_0\rangle_{XYE} |+\rangle_D^{\otimes \Xi} \\
&= 1 - \langle \phi_0 |_{XYE} \langle + |_D^{\otimes \Xi} \left((\mathbb{1} - |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|)_{X_{\mathcal{M}}} (O^H)^\dagger U^\dagger \right)^{q'} \\
&\quad (UO^H (\mathbb{1} - |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|)_{X_{\mathcal{M}}})^{q'} |\phi_0\rangle_{XYE} |+\rangle_D^{\otimes \Xi} \\
&\leq 4 \sum_{i=1}^{q'} \epsilon_{\hat{\mathbf{a}},i}.
\end{aligned}$$

Here, we have used the quantum union bound, Lemma 2, with projectors $P_i = \left((O^H)^\dagger U^\dagger\right)^{i-1} (\mathbb{1} - |\hat{\mathbf{a}}\rangle\langle\hat{\mathbf{a}}|) (UO)^{i-1}$ in the last line. We thus finally get

$$\begin{aligned}
& \Pr_{(x,\pi,w) \leftarrow \langle \mathcal{P}, \mathcal{E} \rangle} [\mathcal{V}(x, \pi) = 1 \wedge (x, w) \notin R] \\
& \leq \delta \sum_{\hat{\mathbf{a}}} \left(\sum_{i=1}^{q'} \left\| |\tilde{\psi}_i\rangle \right\| \right)^2 \\
& \leq 4q' \cdot \delta \sum_{\hat{\mathbf{a}}} \sum_{i=1}^{q'} \epsilon_{\hat{\mathbf{a}}, i} \\
& = 4q'^2 \cdot \delta,
\end{aligned}$$

where we have used the normalization of $|\phi_0\rangle$ in the last line.

3.4 Proof of Zero Knowledge

We prove that the Fischlin transform is zero-knowledge in the quantum random oracle model (QROM). Our proof follows standard techniques for adaptive oracle reprogramming in the QROM [GHHM21]. We begin by defining the zero-knowledge simulator. Let Sim_Σ denote the honest-verifier zero-knowledge simulator for the underlying Σ -protocol.

Definition 11 (Simulator $\text{Sim}_{\text{Fis}}^H$). *The simulator $\text{Sim}_{\text{Fis}}^H$ proceeds as follows.*

Setup. *Sample a random function*

$$\tilde{H} : [k] \times \mathcal{C} \rightarrow \{0, 1\}^l.$$

For each $i \in [k]$, sample $c_i \leftarrow^{\$} \mathcal{C}$ such that $\tilde{H}(i, c_i) = 0^l$. If no such value exists, abort.

Proof Generation. *For each $i \in [k]$, run*

$$(a_i, z_i) \leftarrow \text{Sim}_\Sigma(x, c_i).$$

Set $\mathbf{a} = (a_i)_{i \in [k]}$ and similarly $\mathbf{c}, \mathbf{z}$.

Oracle Programming. *For any query $(\mathbf{a}, i, c, z)$, it answers as follows:*

$$H'(\mathbf{a}, i, c, z) = \begin{cases} \tilde{H}(i, c) & \text{if } V(a_i, c, z) = 1 \text{ and } H \text{ is undefined,} \\ H(\mathbf{a}, i, c, z) & \text{otherwise.} \end{cases}$$

Output. *Define the proof*

$$\pi = (\mathbf{a}, \mathbf{c}, \mathbf{z}).$$

and output (π, H') .

With this simulator in hand, we can prove zero-knowledge.

Theorem 3. *For a Σ -protocol with commitment entropy, the Fischlin transform is zero-knowledge in the QROM.*

Proof. The simulator clearly runs in expected polynomial time.

Hybrid Argument. We prove indistinguishability by a sequence of hybrid experiments.

Hybrid H_0 . The honest prover executes the Fischlin protocol using an unmodified quantum random oracle.

Hybrid H_1 . The prover generates commitments $(a_i)_{i \in [k]}$ honestly. The oracle is reprogrammed on all accepting transcripts with prefix $(\mathbf{a}, i)$ as in Definition 11. Challenges c_i are sampled uniformly, and responses z_i are computed honestly.

By the adaptive reprogramming lemma of [GHHM21] (Lemma 1), H_1 is computationally indistinguishable from H_0 , except with probability negligible in the min-entropy of the commitment vector.

Hybrid H'_1 . As in H_1 , but challenges c_i are sampled conditioned on $\tilde{H}(i, c_i) = 0^l$.

Since $\tilde{H}$ is uniformly random, this conditioning affects the distribution only negligibly. Hence H'_1 is indistinguishable from H_1 .

Hybrid H_2 . As in H'_1 , but responses are generated using the simulator Sim_Σ instead of the honest prover.

By the honest-verifier zero-knowledge property of the Σ -protocol, H_2 is indistinguishable from H'_1 up to advantage at most $k\varepsilon$, where ε is the distinguishing advantage for a single repetition.

Hybrid H_2 corresponds exactly to the output of Sim_{Fis} . Since each transition introduces only negligible distinguishing advantage, the simulator output is computationally indistinguishable from a real proof. This establishes zero knowledge in the QROM. $\square$

4 Acknowledgements

CM thanks Gorjan Alagic for useful discussions. CM and JS acknowledge support by the Independent Research Fund Denmark via a DFF Sapere Aude grant (IM-3PQC, grant ID 10.46540/2064-00034B). We have used AI tools for spell checking, grammar and other English-language-related improvements

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5 Appendix

5.1 Bounding Lemmas

The following Lemmas gives numerical analysis for the bounds.

Lemma 13. *For some constant c , let the challenge space be of size $N = c \cdot 2^l \cdot \log(k)$. Let further $\gamma = 4 \cdot 2^{-l}$, $2^{\frac{1}{c}} \leq k \leq 2^{\frac{2^l}{256 \cdot c}}$ and $l \geq 14$. Then*

$$\varepsilon'' + 2\sqrt{\varepsilon'} + 7\sqrt{\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}} \leq 3 \exp \left[-\frac{k}{128 \cdot c \cdot 2^l \cdot \log k} \right] + 7 \exp \left[-\frac{k}{8 \cdot 2^l} \right] = \text{negl}(k).$$

Proof. Substituting the formulas for the parameters we get

$$\begin{aligned}\bar{\varepsilon} &:= \varepsilon'' + 2\sqrt{\varepsilon'} + 7\sqrt{\sqrt{\varepsilon_{\gamma, (1-\gamma)k}}} \\ &\leq \exp\left[-\frac{\delta^2\mu}{3}\right] + 2\exp\left[-\frac{\delta'^2\mu'}{2^{l+1}}\right] + 7\exp\left[-(\gamma - 2 \cdot 2^{-l})(1-\gamma)k/8\right]\end{aligned}\quad (25)$$

using Lemma 14 below.

We simplify

$$\bar{\varepsilon} \leq \exp\left[-\frac{\delta^2\mu}{3}\right] + 2\exp\left[-\frac{\delta'^2\mu'}{2^{l+1}}\right] + 7\exp\left[-2^{-l}k/8\right]$$

where we have used $1 - \gamma \geq 1/2$.

We bound

$$\begin{aligned}\underline{\mu}' &= 2^{-l}k \left[N - 1 - \gamma \left(1 + 2^{2+l} + \log\left(\frac{4\gamma}{\log k}\right) \right) - 2^{2+l/2}\sqrt{N\gamma} \right] \\ &= \mu - 2^{-l}k \left[1 + \gamma \left(1 + 2^{2+l} + \log\left(\frac{4\gamma}{\log k}\right) \right) + 2^{2+l/2}\sqrt{N\gamma} \right] \leq \mu\end{aligned}\quad (26)$$

as $\log\left(\frac{4\gamma}{\log k}\right) \geq 4 - 2l \geq -4 \cdot 2^l$ by the assumptions $k \leq 2^{2^l}$ and $l \geq 14$.

For simplicity, we set $\delta\mu = \delta'\underline{\mu}'$. We thus get

$$\begin{aligned}\delta'^2\underline{\mu}' &= \frac{\delta^2\mu^2}{\underline{\mu}'} \\ &\geq \delta^2\mu.\end{aligned}\quad (27)$$

using Eq. (26).

We have $\eta = \mu(1+\delta)$ (from applying Lemma 5) and $\eta - (1-2\gamma)k = \underline{\mu}'(1-\delta')$ (see Lemma 11). We thus get

$$\begin{aligned}\mu(1+\delta) - (1-2\gamma)k &= \underline{\mu}'(1-\delta') \\ \Leftrightarrow \delta &= \frac{(1-8 \cdot 2^{-l})k - (\mu - \underline{\mu}')}{2\mu}.\end{aligned}\quad (28)$$

We bound

$$\begin{aligned}\mu - \underline{\mu}' &= 2^{-l}k \left[1 + \gamma \left(1 + 2^{2+l} + \log\left(\frac{4\gamma}{\log k}\right) \right) + 2^{2+l/2}\sqrt{N\gamma} \right] \\ &\leq 2^{-l}k \left[1 + 4 \cdot 2^{-l} (1 + 4 \cdot 2^l) + 8\sqrt{N} \right] \\ &\leq 2^{-l/2}k \left[21 \cdot 2^{-l/2} + 8\sqrt{c \cdot \log k} \right]\end{aligned}$$

We can use the assumption on k ,

$$8\sqrt{c \cdot \log k} \leq 8\sqrt{\frac{2^l}{256}} \leq \frac{1}{2}2^{l/2},$$

so we get

$$\begin{aligned}\mu - \underline{\mu}' &\leq 2^{-l/2}k \left[21 \cdot 2^{-l/2} + 8\sqrt{c \cdot \log k} \right] \\ &\leq \frac{3k}{4},\end{aligned}$$

using $l \geq 14$. Plugging into Eq. (28) we get

$$\begin{aligned}\delta &\geq \frac{\left(\frac{1}{4} - 8 \cdot 2^{-l}\right)k}{\mu} \\ &\geq \frac{1}{8c \log k}.\end{aligned}$$

Here we have used $8 \cdot 2^{-l} \leq 2^{-11} \leq 1/8$ due to the assumption $l \geq 14$. We can now use this to get

$$\begin{aligned}\delta^2 \mu &\geq \frac{2^{-l}kN}{64c^2 \log^2 k} \\ &= \frac{2^{-l}k \cdot c \cdot 2^l \cdot \log k}{64c^2 \log^2 k} \\ &= \frac{k}{64c \log k}.\end{aligned}$$

Finally, we obtain

$$\begin{aligned}\bar{\varepsilon} &\leq \exp\left[-\frac{\delta^2 \mu}{3}\right] + 2 \exp\left[-\frac{\delta'^2 \underline{\mu}'}{2^{l+1}}\right] + 7 \exp[-2^{-l}k/8] \\ &\leq \exp\left[-\frac{\delta^2 \mu}{3}\right] + 2 \exp\left[-\frac{\delta^2 \mu}{2^{l+1}}\right] + 7 \exp[-2^{-l}k/8] \\ &\leq 3 \exp\left[-\frac{\delta^2 \mu}{2^{l+1}}\right] + 7 \exp[-2^{-l}k/8] \\ &\leq 3 \exp\left[-\frac{k}{128 \cdot c \cdot 2^l \cdot \log k}\right] + 7 \exp[-2^{-l}k/8].\end{aligned}$$

□

We prove the deferred lemma.

Lemma 14. *For some constant c , let the challenge space be of size $N = c \cdot 2^l \cdot \log(k)$. Let further $\gamma = 4 \cdot 2^{-l}$, $2^{\frac{1}{c}} \leq k \leq 2^{\frac{2^l}{256 \cdot c}}$ and $l \geq 14$. Then*

$$\exp\left(-\frac{\delta'^2 \underline{\mu}'^2}{2m}\right) \leq \exp\left(-\frac{\delta'^2 \underline{\mu}'}{2^l}\right).$$

Proof. We bound

$$\begin{aligned}\exp\left(-\frac{\delta'^2 \underline{\mu}'^2}{2m}\right) &\leq \exp\left(-\frac{\delta'^2 \underline{\mu}'^2}{2kN}\right) \\ &\leq \exp\left(-\frac{\delta'^2 \underline{\mu}'}{2^l}\right),\end{aligned}$$

where the first inequality is straightforward from $m = k(N-1) \leq kN$, and the second inequality is obtained using the definition of $\underline{\mu}'$ and the assumptions on the parameters,

$$\begin{aligned}\frac{\underline{\mu}'}{N} &= \frac{2^{-l}k \left[N - 1 - \gamma \left(1 + 4 \cdot 2^l - \log \left(\frac{\log k}{4\gamma} \right) \right) - 4\sqrt{2^l \gamma N} \right]}{kN} \\ &\geq 2^{-l} \left[1 - \frac{(1 + \gamma(1 + 4 \cdot 2^l) + 4\sqrt{2^l \gamma N})}{N} \right] \\ &\geq 2^{-l} \left[1 - \frac{(1 + 4 \cdot 2^{-l}(1 + 4 \cdot 2^l) + 4\sqrt{4N})}{N} \right] \\ &\geq 2^{-l} \left[1 - \frac{(21 + 8\sqrt{N})}{N} \right] \\ &\geq 2^{-l} \left[1 - \frac{29}{\sqrt{c \log k} \cdot 2^l} \right] \\ &\geq 2^{-l} \left[1 - \frac{29}{\sqrt{2^l}} \right] \\ &\geq 2^{-l} [1 - 29 \cdot 2^{-7}] \\ &\geq \frac{1}{2} 2^{-l}\end{aligned}$$

□